Learning Objectives

By the end of this lecture, you should be able to:

1. Understand common statistical model selection approaches, including AIC, model averaging, and stepwise regression, and recognise their limitations and potential pitfalls.

2. Understand the principles of causal model building, including directed acyclic graphs (DAGs), d-separation, and Structural Equation Modelling (SEM).

3. Appreciate the difference between predicting relationships and understanding causal relationships.

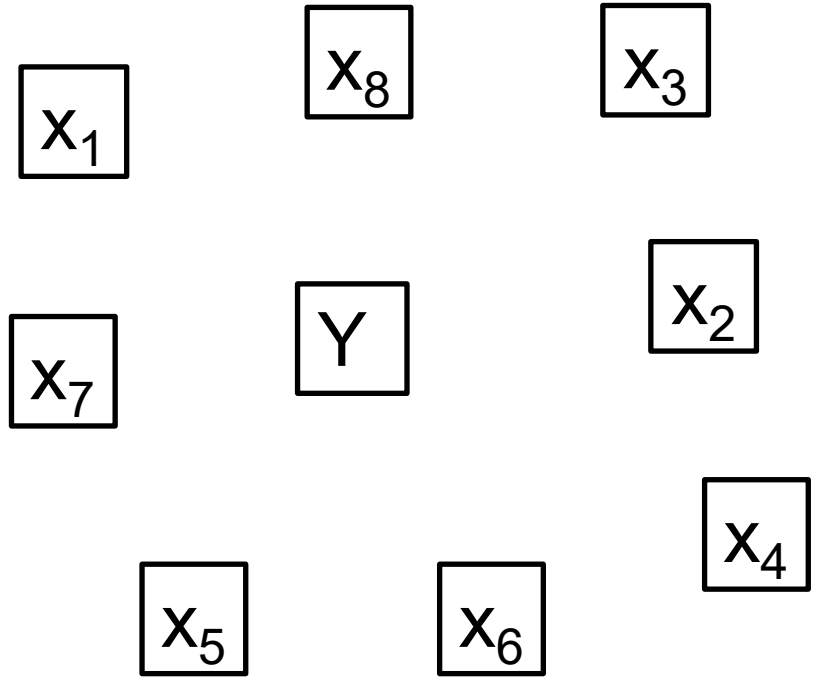
Please go to this link, everything you need to do is explained there,

https://pad.riseup.net/p/ES3307_Tharaka-keep

OK, Our data is in, Now what to do with it?



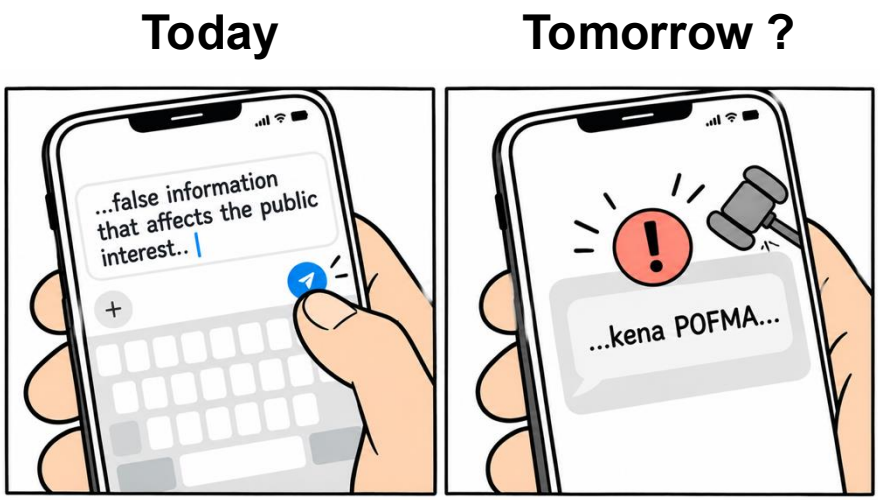
But Which Predictors? All? Some? Which?!?!?!?!



Information Theory

Information: Reduction in uncertainty caused by leaning an outcome (i.e., saying that when something is more uncertain than something else).

Information theory: A framework deriving rigorous metric for uncertainty (e.g., AIC = Akaike [An] Information Criteria).





Hirotugu Akaike

How many parameters does it take to fit an elephant?

- Lower AIC means less information is lost by a model (i.e., most predictive model).

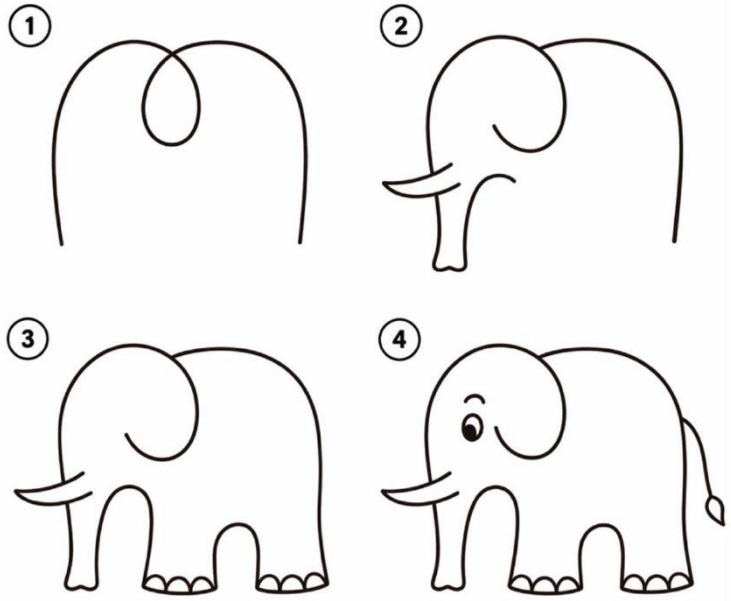
$$AIC = -2 \log(L(\hat{\theta} \mid data)) + 2K$$

K = Number of Parameters

- We want to balance the idea of perfect fit with generality.







Model Mis-selection

Let's open Script_1 and run following codes

```
## stepwise regression ----------------------------------------------

# backward
F1 <- lm(math ~ ., data = student_df)
F1_step_back <- step(F1, direction = "backward")

# forward
F2 <- lm(math ~ 1, data = student_df)
F2_step_forward <- step(F2, direction = "forward", scope = formula(F1))

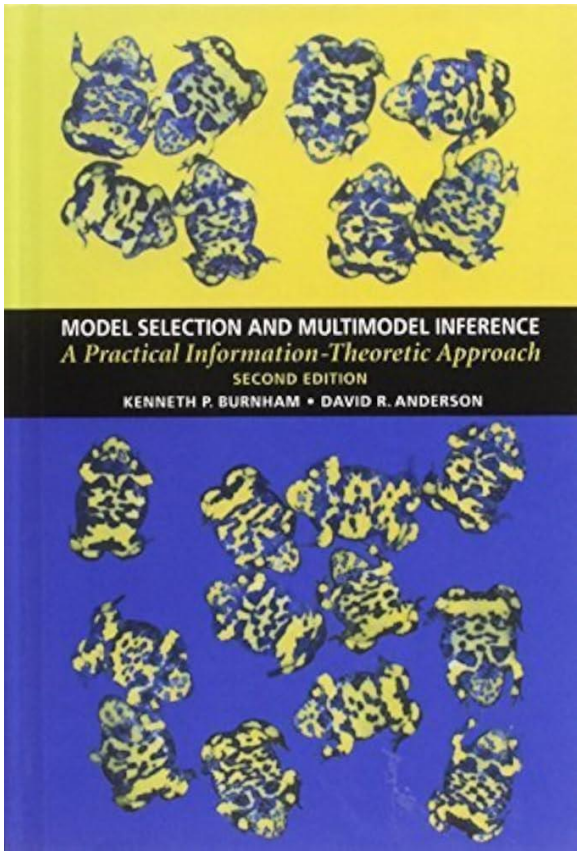
# both backward and forward
F3_step_both <- step(F1, direction = 'both')
# or
F4_step_both <- step(F2, direction = 'both', scope = formula(F1))
```

Model Mis-selection

Model averaging

AIC diff	support for equivalency of models
0-2	substantial
4-7	weak
> 10	none

Burnham, K.P. and Anderson, D.R. 2002. Model Selection and Multimodel Inference. Springer Verlag. (second edition), p 70.



Model Averaging

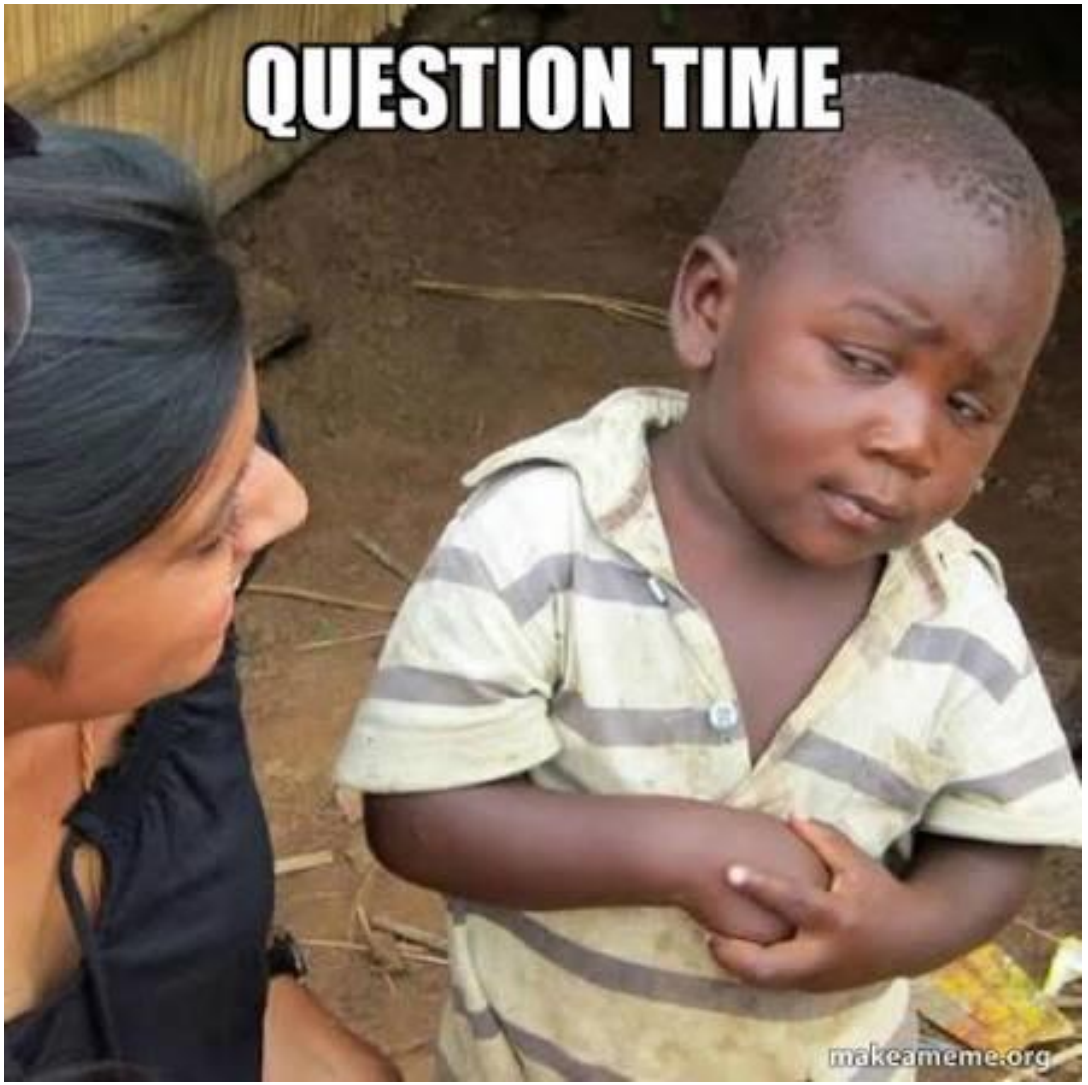
Let's open Script_1 and run following codes

```
> # I use swiss datset to illustrate this
> library(datasets)
> data(swiss)
> head(swiss)
             Fertility Agriculture Examination Education Catholic
Courtelary        80.2        17.0          15        12     9.96
Delemont          83.1        45.1           6         9    84.84
Franches-Mnt      92.5        39.7           5         5    93.40
Moutier           85.8        36.5          12         7    33.77
Neuveville        76.9        43.5          17        15     5.16
Porrentruy        76.1        35.3           9         7    90.57
             Infant.Mortality
Courtelary               22.2
Delemont                 22.2
Franches-Mnt             20.2
Moutier                  20.3
Neuveville               20.6
Porrentruy               26.6
>
```

For more details about the dataset, https://stat.ethz.ch/R-manual/R-devel/library/datasets/html/swiss.html

```
library(MuMIn)
Q1 <- lm(Fertility ~ ., data = swiss, na.action = "na.fail")
mmQ1 <- dredge(Q1)

mmQ2_avg <- model.avg(mmQ1, subset = delta < 4)
summary(mmQ2_avg)
```


QUESTION TIME
makeameme.org

Model Mis-selection

M1 <- lm(Fertility ~ Agriculture + Education + Catholic, data = swiss)

summary()

```
> # so let's say I fit a model like below
> M1 <- lm(Fertility ~ Agriculture + Education + Catholic, data = swiss)
> summary(M1)

Call:
lm(formula = Fertility ~ Agriculture + Education + Catholic,
    data = swiss)

Residuals:
    Min      1Q  Median      3Q     Max
-15.178  -6.548   1.379   5.822  14.840

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 86.22502    4.73472  18.211  < 2e-16 ***
Agriculture -0.20304    0.07115  -2.854  0.00662 **
Education   -1.07215    0.15580  -6.881 1.91e-08 ***
Catholic     0.14520    0.03015   4.817 1.84e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.728 on 43 degrees of freedom
Multiple R-squared:  0.6423,    Adjusted R-squared:  0.6173
F-statistic: 25.73 on 3 and 43 DF,  p-value: 1.089e-09
```

anova()

```
> # this is sequential anova or Type I sums of squares
> anova(M1)
Analysis of Variance Table

Response: Fertility
            Df  Sum Sq Mean Sq F value    Pr(>F)
Agriculture  1  894.84  894.84  14.984 0.0003635 ***
Education    1 2329.85 2329.85  39.014 1.603e-07 ***
Catholic     1 1385.39 1385.39  23.199 1.842e-05 ***
Residuals   43 2567.88   59.72
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

Summary () vs anova()

Key Differences

- `summary()` **(Regression Output)**
  - **Test type:** Performs individual t-tests (or z-tests) for each model coefficient.
  - **What it shows:** The effect of each predictor level *relative to a reference level* (for categorical variables) or the slope (for continuous variables).
  - **Interactions:** Displays individual slope or intercept changes for specific combination terms.
  - **p-values:** Unadjusted p-values for each specific coefficient term. Stack Exchange +2
- `anova()` **(Analysis of Variance Table)**
  - **Test type:** Performs F-tests comparing nested models or evaluating overall main and interaction effects.
  - **What it shows:** The overall significance of a factor as a whole (combining all its levels), rather than comparing individual levels to a baseline.
  - **Order sensitivity:** Base `anova()` uses sequential (Type I) sums of squares by default, meaning the order of variables in your model formula changes the output. *(Note: To get marginal/Type III tests similar to other software, you can use* `car::Anova(..., type = 3)` *.)* Stack Exchange +6

This is an AI summary

anova()

M1 <- lm(Fertility ~ Agriculture + Education + Catholic, data = swiss)

```
> # this is sequential anova or Type I sums of squares
> anova(M1)
Analysis of Variance Table

Response: Fertility
            Df  Sum Sq Mean Sq F value    Pr(>F)
Agriculture  1  894.84  894.84  14.984 0.0003635 ***
Education    1 2329.85 2329.85  39.014 1.603e-07 ***
Catholic     1 1385.39 1385.39  23.199 1.842e-05 ***
Residuals   43 2567.88   59.72
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

```
# this is what is happening in here
sum(residuals(lm(Fertility ~ 1, data = swiss))^2) - sum(residuals(lm(Fertility ~ Agriculture, data =
swiss))^2) #see it's 894.8391

sum(residuals(lm(Fertility ~ Agriculture, data = swiss))^2) - sum(residuals(lm(Fertility ~
Agriculture + Education, data = swiss))^2) #see it's 2329.85

sum(residuals(lm(Fertility ~ Education + Agriculture, data = swiss))^2) - sum(residuals(lm(Fertility
~ Agriculture + Education + Catholic, data = swiss))^2)#see it's 1385.386

sum(residuals(lm(Fertility ~ Agriculture + Education + Catholic, data = swiss))^2) #see it's 2567.88
(note: this is RSS)
```

So be Careful!

M1 <- lm(Fertility ~ Agriculture + Education + Catholic, data = swiss)

M2 <- lm(Fertility ~ Education + Agriculture + Catholic, data = swiss)

```
> # so if I now do sequential anova, you see I get different results
> anova(M1)
Analysis of Variance Table

Response: Fertility
            Df  Sum Sq Mean Sq F value    Pr(>F)
Agriculture  1  894.84  894.84  14.984 0.0003635 ***
Education    1 2329.85 2329.85  39.014 1.603e-07 ***
Catholic     1 1385.39 1385.39  23.199 1.842e-05 ***
Residuals   43 2567.88   59.72
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> anova(M2)
Analysis of Variance Table

Response: Fertility
            Df Sum Sq Mean Sq F value    Pr(>F)
Education    1 3162.7  3162.7 52.9607 5.114e-09 ***
Agriculture  1   62.0    62.0  1.0376    0.3141
Catholic     1 1385.4  1385.4 23.1987 1.842e-05 ***
Residuals   43 2567.9    59.7
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```



Anova() vs drop1()

M1 <- lm(Fertility ~ Agriculture + Education + Catholic, data = swiss)

drop1(M1, scope = ~ Agriculture + Education + Catholic, test = "F")

Anova()

```
> # now look at Anova (default is Type II tests)  ------------------------
> library(car)
Loading required package: carData
> Anova(M1)
Anova Table (Type II tests)

Response: Fertility
             Sum Sq Df F value    Pr(>F)
Agriculture  486.28  1   8.143  0.006623 **
Education   2827.94  1  47.355 1.911e-08 ***
Catholic    1385.39  1  23.199 1.842e-05 ***
Residuals   2567.88 43
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

drop1()

```
> drop1(M1, scope = ~ Agriculture + Education + Catholic, test = "F")
# this is the same as car:: Anova
Single term deletions

Model:
Fertility ~ Agriculture + Education + Catholic
            Df Sum of Sq    RSS    AIC F value    Pr(>F)
<none>                  2567.9 196.03
Agriculture  1    486.28 3054.2 202.18   8.143  0.006623 **
Education    1   2827.94 5395.8 228.93  47.355 1.911e-08 ***
Catholic     1   1385.39 3953.3 214.31  23.199 1.842e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

```r
sum(residuals(lm(Fertility ~ Education + Catholic, data = swiss))^2) -
sum(residuals(lm(Fertility ~ Agriculture + Education + Catholic, data =
swiss))^2) #see it's 486.2846

sum(residuals(lm(Fertility ~ Agriculture + Catholic, data = swiss))^2) -
sum(residuals(lm(Fertility ~ Agriculture + Education + Catholic, data =
swiss))^2) #see it's 2827.941

sum(residuals(lm(Fertility ~ Agriculture + Education, data = swiss))^2) -
sum(residuals(lm(Fertility ~ Agriculture + Education + Catholic, data =
swiss))^2) #see it's 1385.386
```

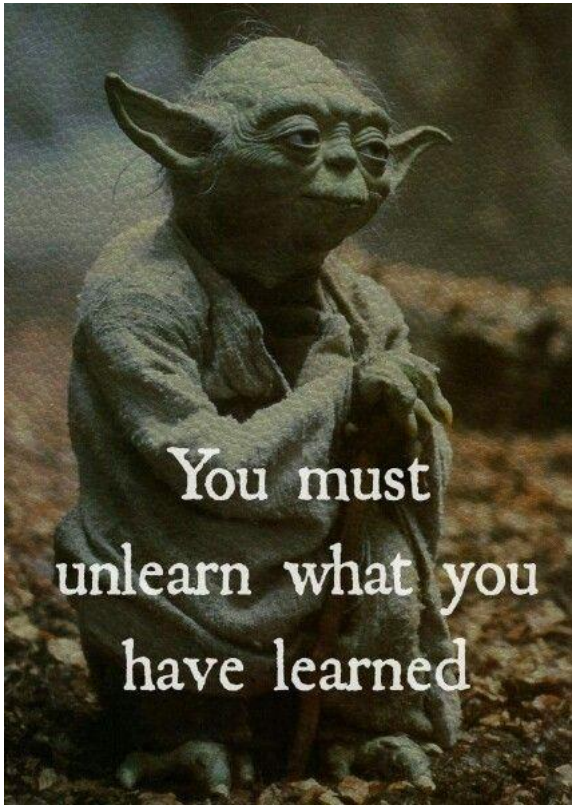
Model Criticism and Simplification

- It is easy to “fall in love” with a particular model, but it is important to remember:
  - All models are wrong.
  - Some models are better than others.
  - We can never be certain which is the correct model.
  - The simpler the model, the better it is.



In Summary

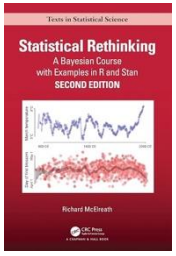
- Model comparison is not causal inference.
- Stop testing, start thinking.



Causal Inference

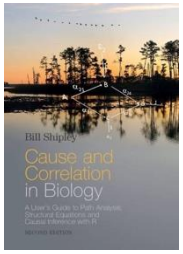
Credits:





Richard McElreath





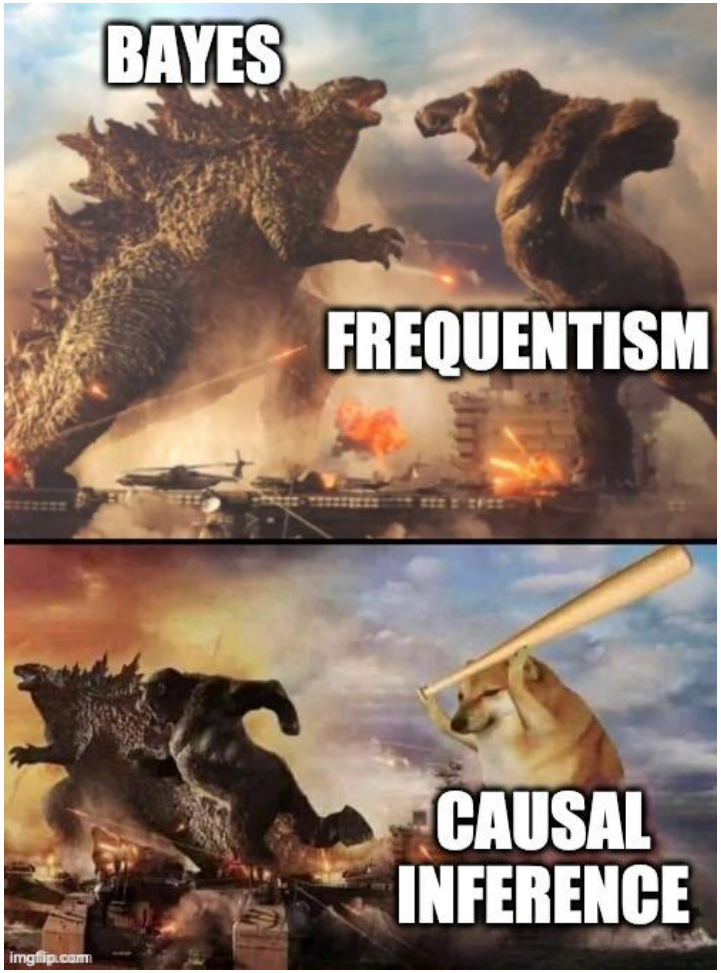
Bill Shipley



Jarrett Byrnes



Jonathan Lefcheck



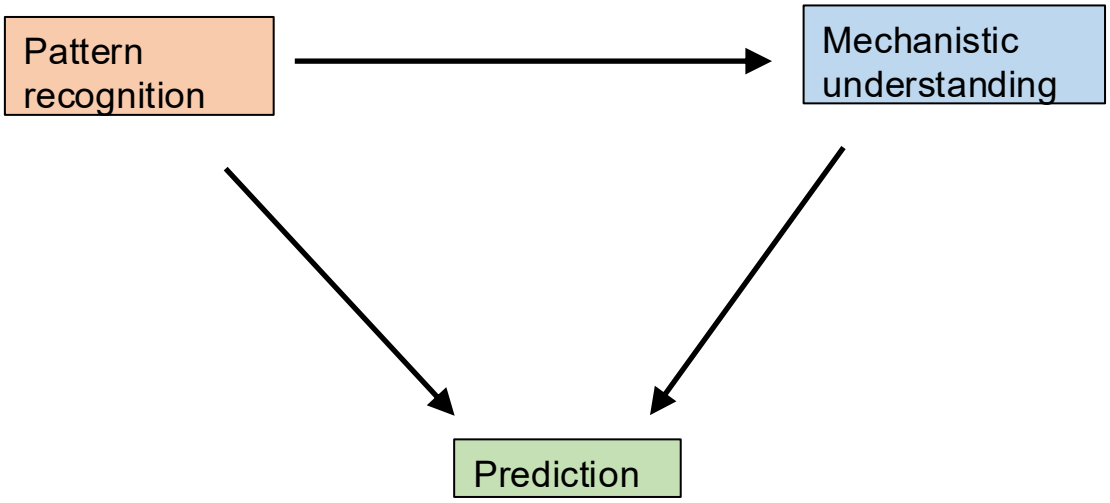
What is Causal Inference?

- Causal inference is more than learning an **association** (i.e., mutual information among variables).

Two concepts:

- Causal inference is **prediction of intervention**.
- Causal inference is **missing data imputation**.

Goals of Science



Causal Imputation



- Knowing a cause means being able to construct unobserved counterfactual outcomes.
- *What if I had done something else?*

Today's Menu

Part 1. Causal design:

Thinking about causal model structure to build valid inference.

Part 2. Building causal models:

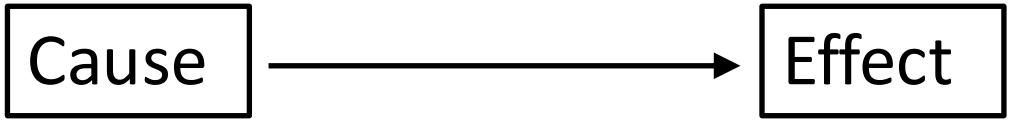
Directed Acyclic Graph (DAG),

D-separation, and

Structural Equation Model (SEM).



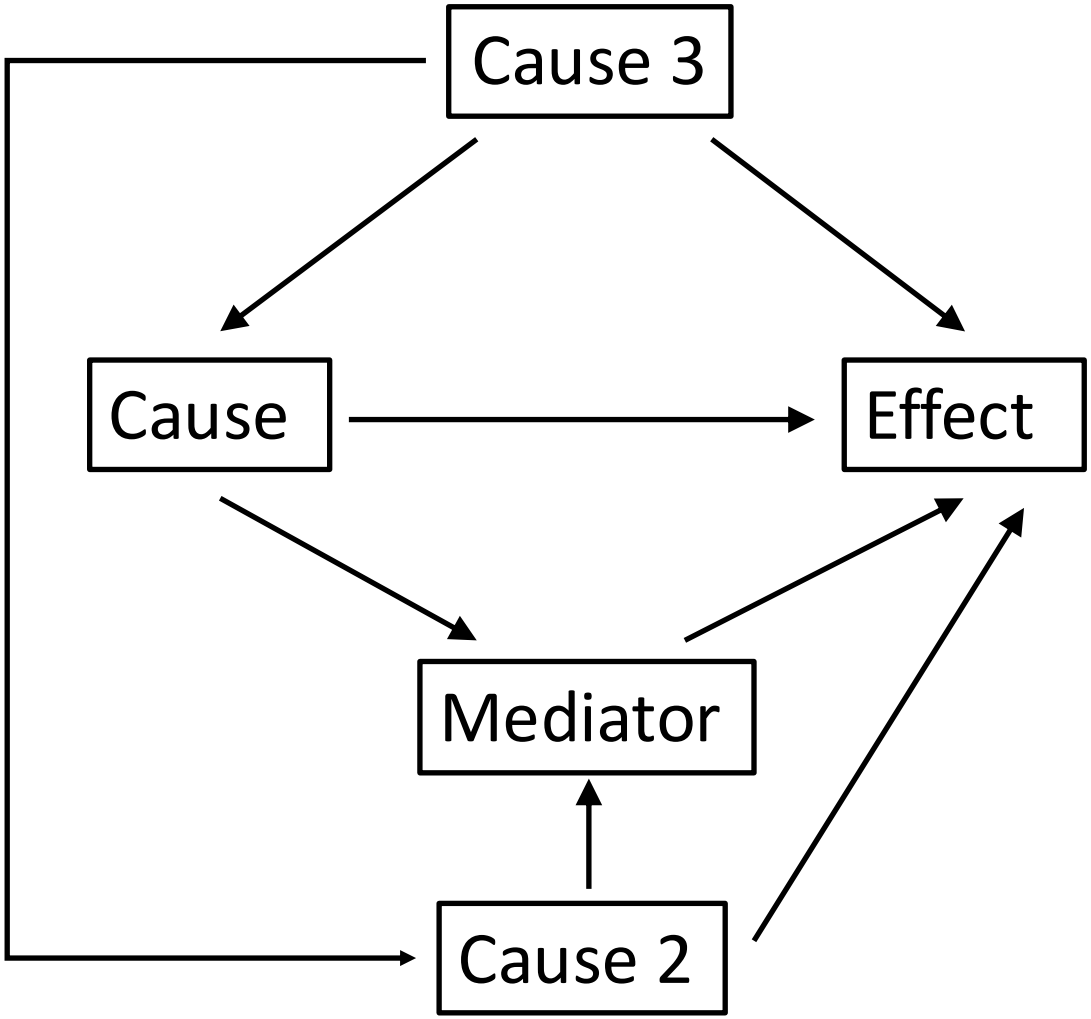
What We Want to Evaluate



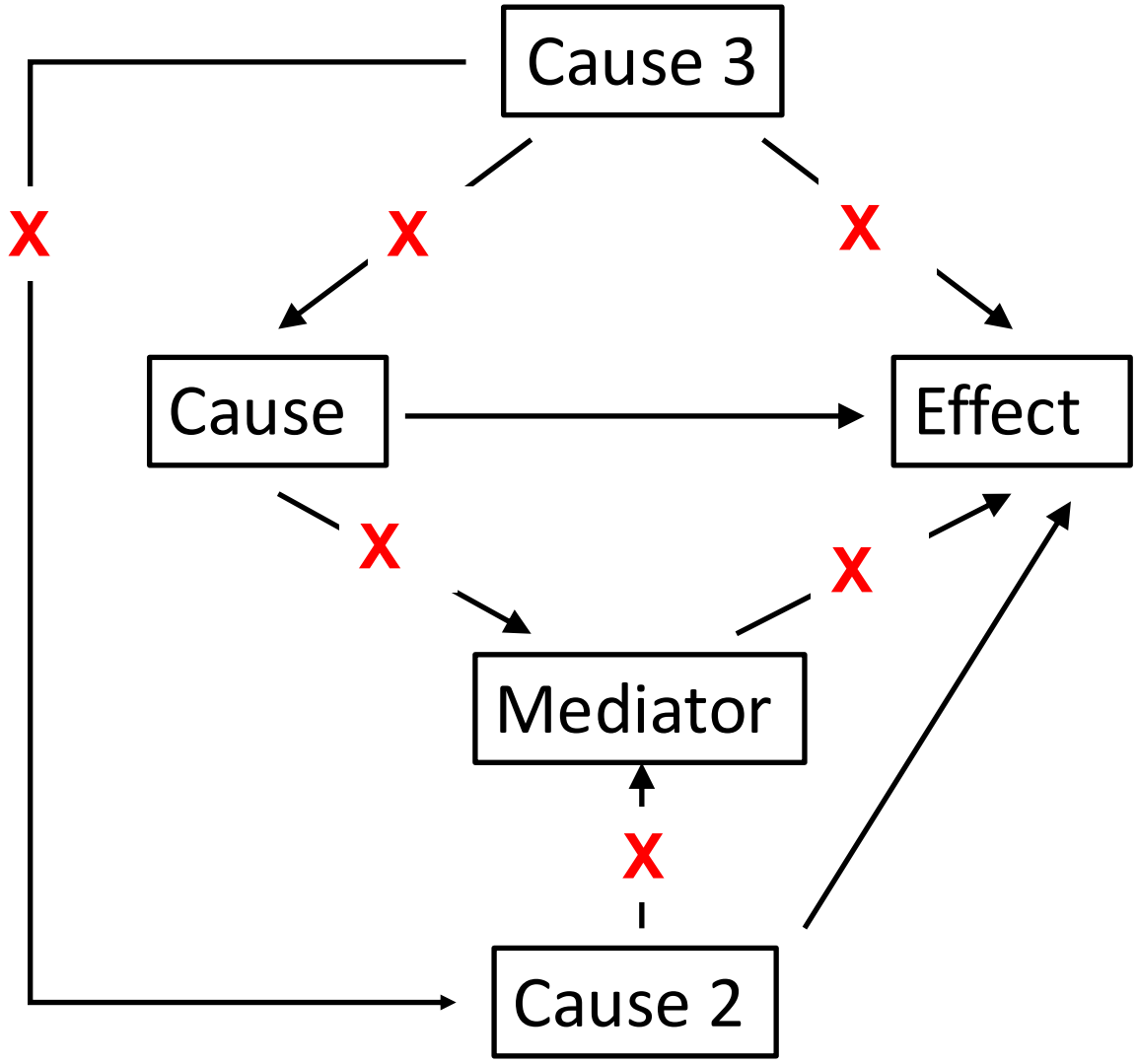
- Exogenous variable
- Independent variable
- Explanatory variable
- Predictor

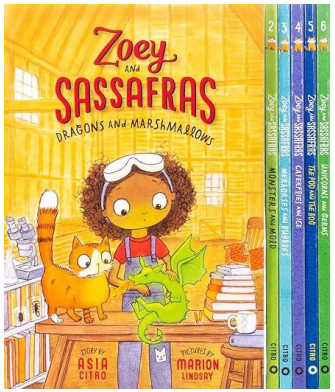
- Endogenous variable
- Dependent variable
- Response variable

But this is the World

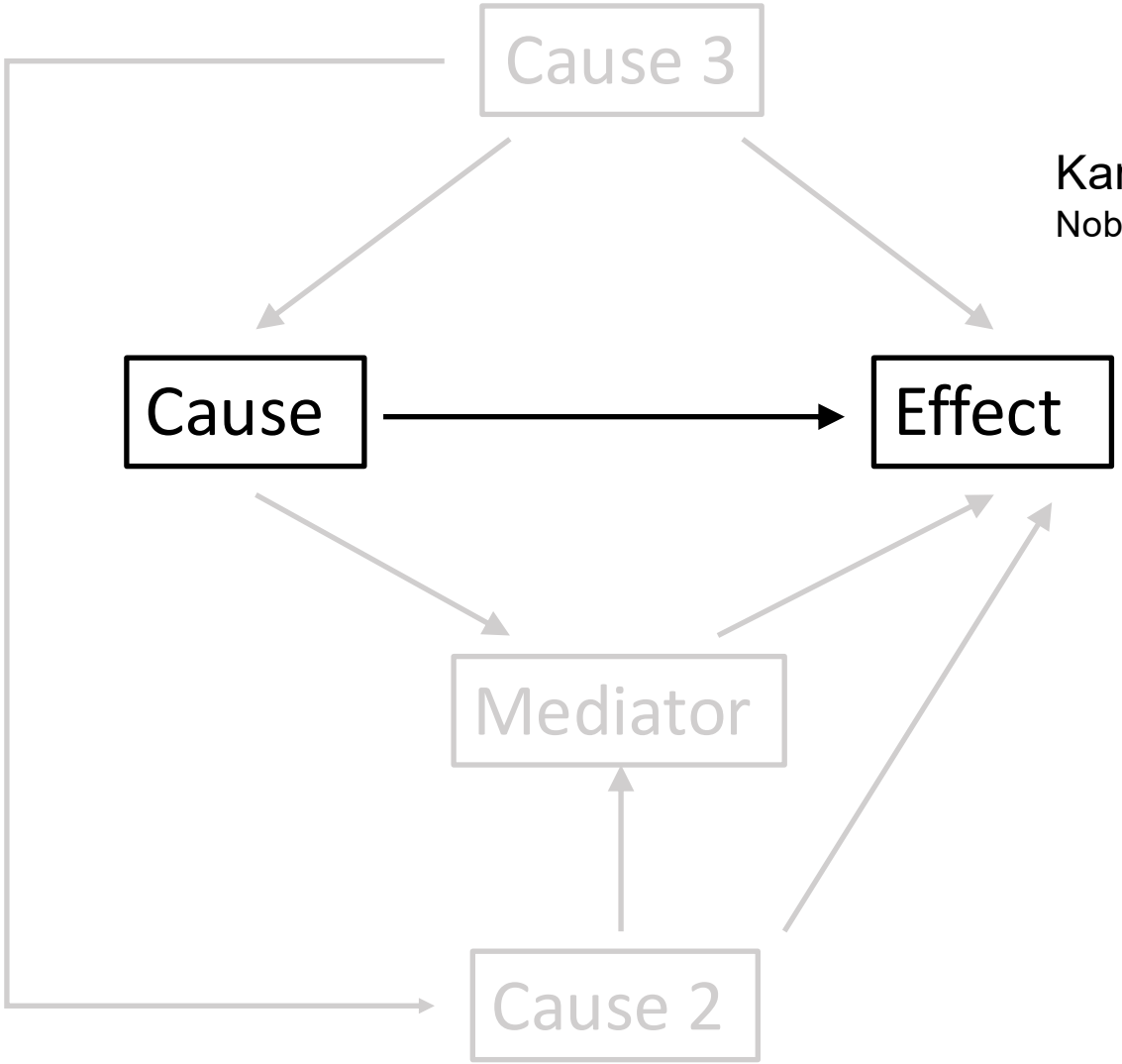


Intervention: Experiments! What can we learn?





Experiments: Manipulate Cause of Interest



Karl Deisseroth –
Nobel Prize winner 2026

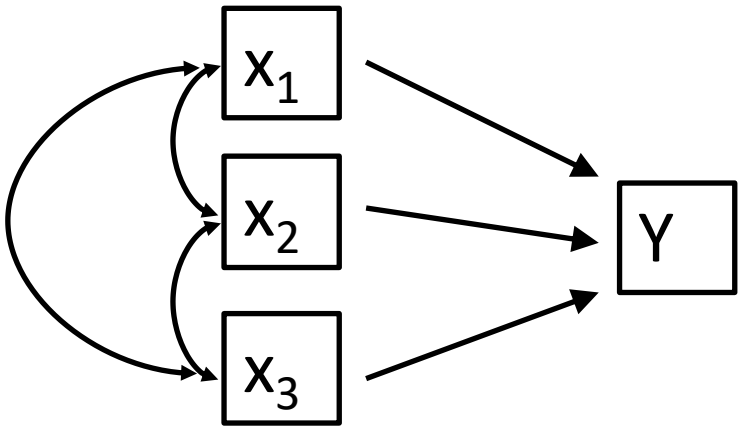






https://www.youtube.com/watch?v=bggKUbIN0Ww&t=624s

Can We Think of Multiple Regression from a Causal Standpoint?



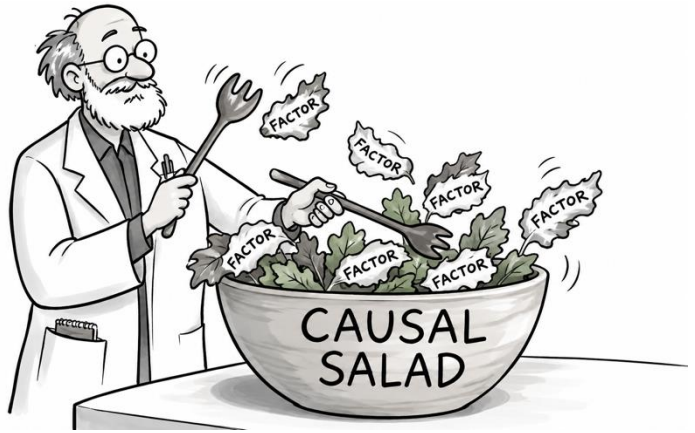
- We estimate the effect of exogenous variables ***controlling*** for all others
- Covariances implied
- Not controlling for the right variables = bad inference
- Controlling for the wrong variables = bad inference

Causal Salad

- Ingredients: Vague query, some relevant variables.
- Add all the variables and let multiple regression sort it out.
- Pretend there is no confounding.
- Pretend there are is no measurement error.
- Pretend pattern of missing data is of no concern.
- Pretend AIC/p-values will pick the scientifically correct model.

Observational studies with lots of variables: IT Approach – AIC's

- Multiple regression needs a different approach.
- **First check collinearity**
- Collinear variables – explain the same variance.
- Very correlated variables e.g. with a correlation coefficient of >0.7 should probably be removed from the model (i.e. select only 1 of the correlated variables).
- Can use *pairs(dataset)* to check correlations and spread of data.
- Use *vif()* to check correlations – variance inflation factor



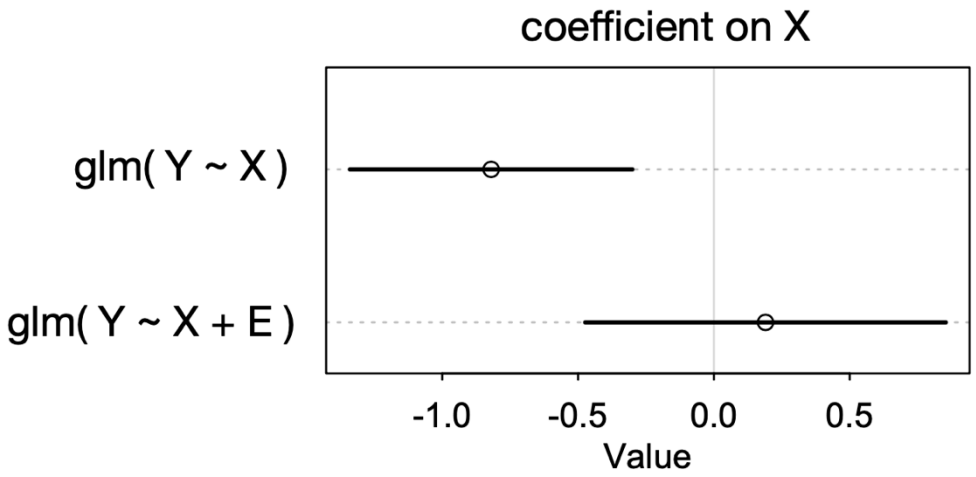
Simulation example: Do students from developing countries face discrimination in PhD scholarship selection?

- 500 students apply for PhD scholarships in two funding schemes (**E**).
- **E: Scholarship type** (i.e., highly competitive international scholarships vs targeted development scholarships).
- **X:** Applicant's country category (i.e, developing or developed country)
- **Y: Scholarship outcome** (i.e., awarded or rejected).
- **Simulation:** Within each scholarship scheme, there is **NO** discrimination based on country category.

- Consider two regressions (Logistic):

Y ~ X X: Applicant's country

Y ~ X + E E: Scholarship type



- Why does E make false evidence of discrimination vanish?

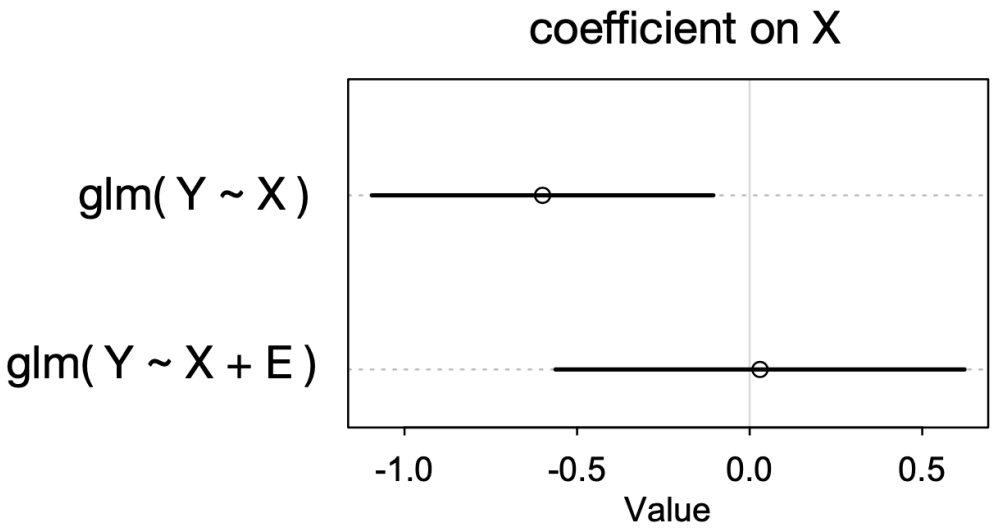
Simulation example: Do students from developing countries face discrimination in PhD scholarship selection?

- 500 students apply for PhD scholarships in two funding schemes (**E**).
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- **X:** Applicant's country category (i.e, developing or developed country)
- **Y: Scholarship outcome** (i.e., awarded or rejected).
- **Simulation:** Within each scholarship scheme, there **IS** discrimination based on country category.

- Consider two regressions (Logistic):

Y ~ X X: Applicant's country

Y ~ X + E E: Scholarship type



- Why does E make discrimination vanish, even though there is discrimination in the simulation?

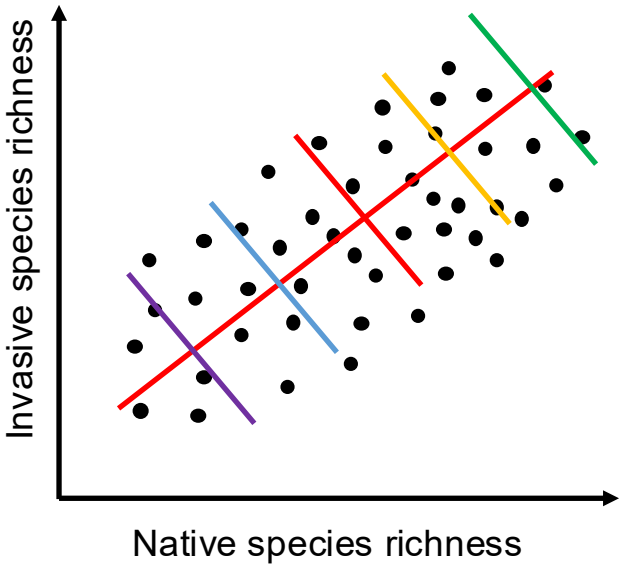
What we can learn from this ?

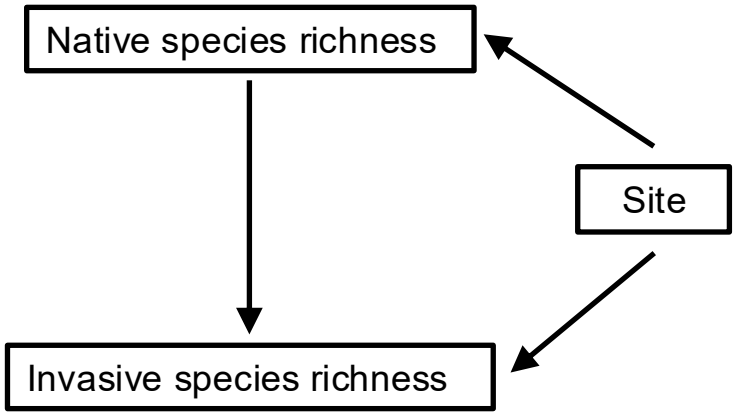
- In real data, we do not know the truth
- Change of coefficient could be revelation or deception

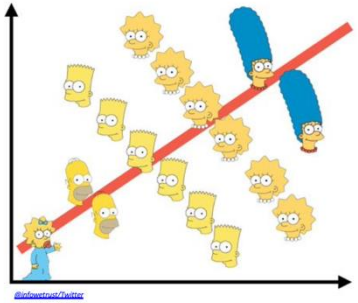


Why use multiple predictors: Simpson's paradox

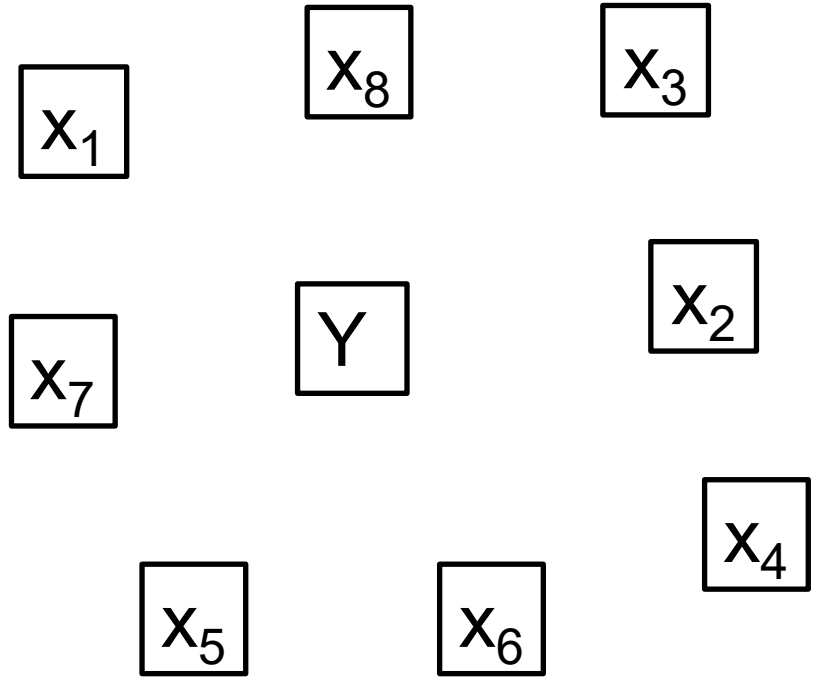
- Classic Problem: Does having more native species hinder invasive species success?







But Which Predictors? All? Some? Which?!?!?!?!



Why not to use all predictors

1. Increased SE on all coefficients

2. Spurious correlations might give false signals

3. Including variables that can obfuscate the signal of causation due to causal graph structure.

Causal Design



Causal Design

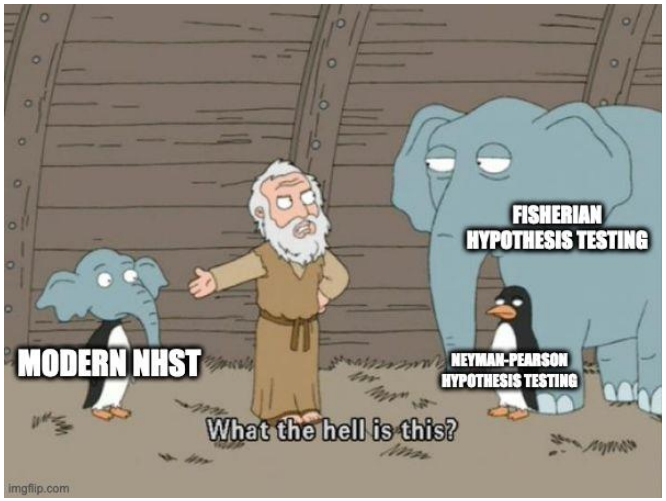
• Causal inference requires a model outside of the statistical model.

Step 1: Make a causal model

- In simplest case, can represent these functions with arrows that connect variables (**DAG**: Directed Acyclic Graphs).

- There is no method for making causal models other than science. There is no method to science other than honest anarchy.

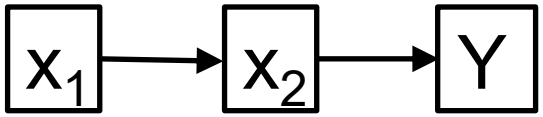
Step 2: Use the model to design data collection and statistical procedures.



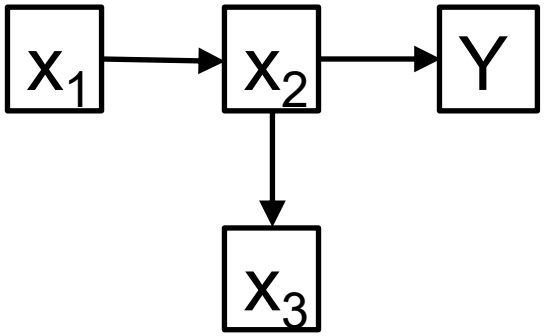
Modern NHST: Fisher's p-values, Neyman–Pearson's decision rules, and a lot of confusion about what statistical significance actually means.

Elemental paths in causal graphs (and Why to use SEM)

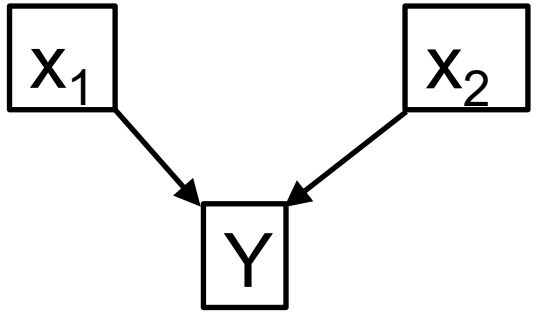
1. The Chain/Pipe



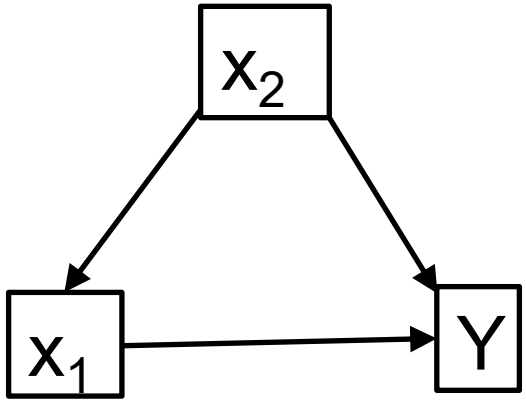
2. The Descendant



3. The Collider



4. The Fork



Chain/Pipe of foolish inference

Consider the following:

- You are following a cohort of patients with tumors to see if tumor size influences buildup of blood toxins. You measure differences in tumor sizes at t0, and then again at t1, when you take blood samples.

- Which tumor size(s) do you use as predictors of blood toxin concentrations? Initial, final, or both?

Truth

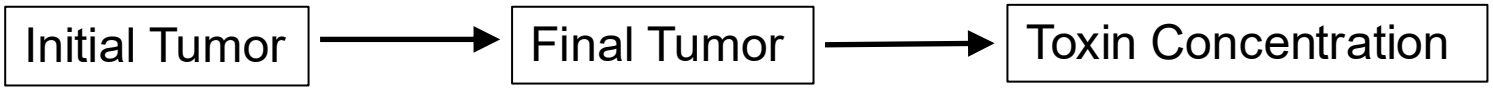


Causal model



Consider the consequences

Truth

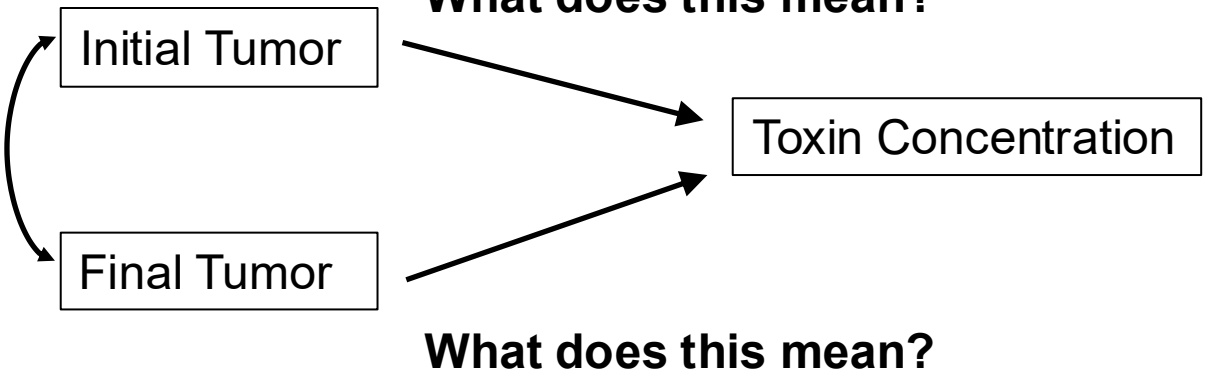


Regression 1 **Net effect, ignoring mediator**



Regression 2

Adjust effects holding mediator or initial constant



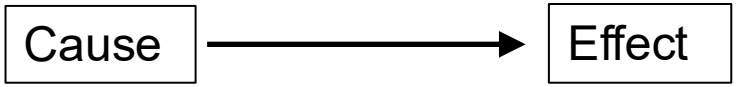
1. The Chain/Pipe

Causal model



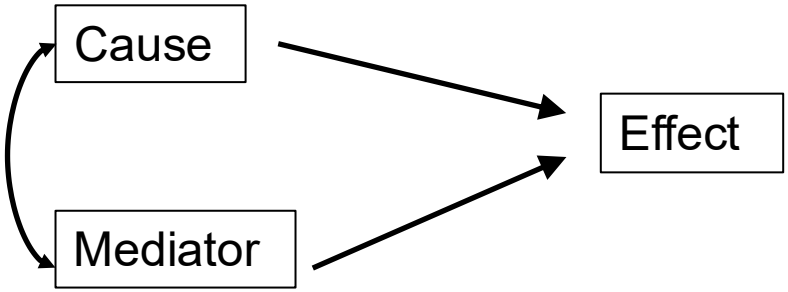
• **Mediator blocks cause**

Regression 1



• **Ignoring Mediator,**
Cause and Effect associated

Regression 2

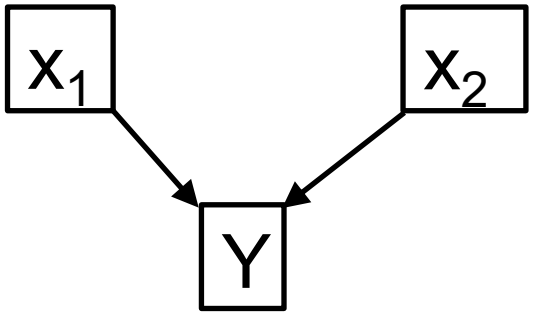


• **Conditioned by Mediator,**
Cause and Effect NOT associated

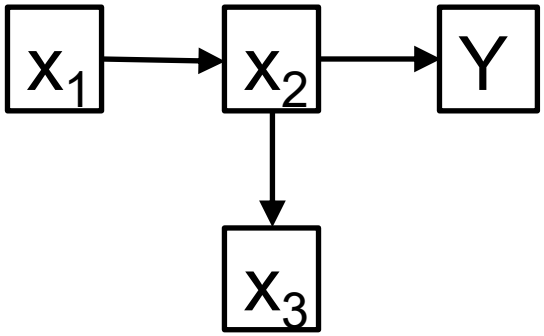
Elemental paths in causal graphs (and Why to use SEM)

1. The Chain/Pipe

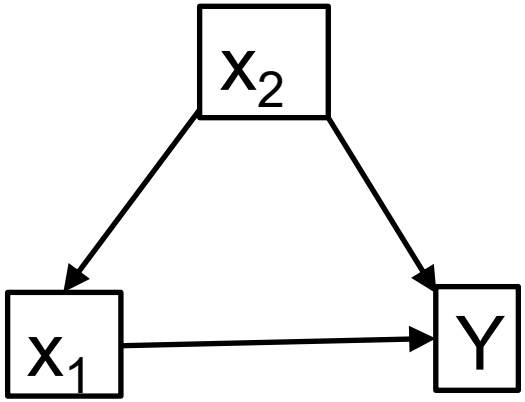
3. The Collider



2. The Descendant



4. The Fork

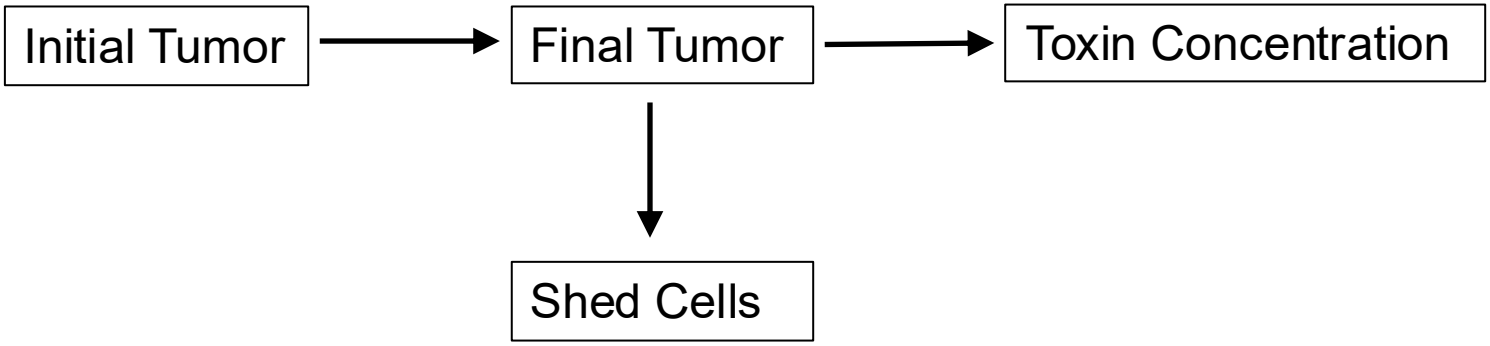


Will a child set us free?

Consider the following:

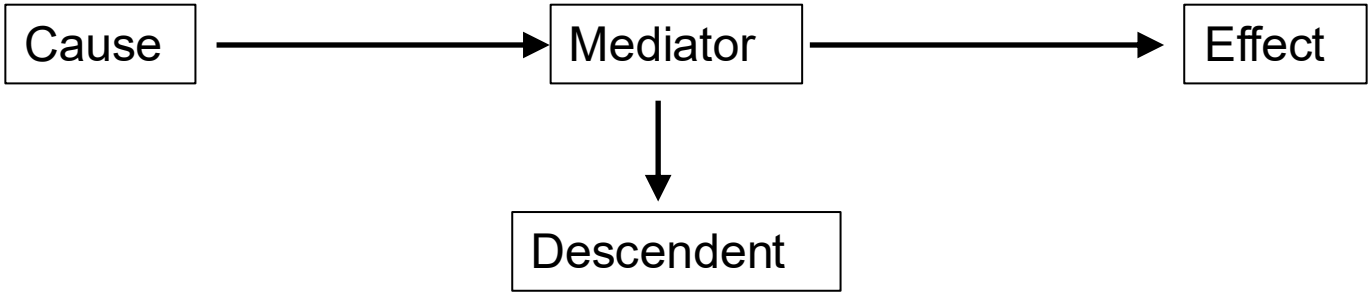
What if we knew the final tumor was shedding cells, and we had their concentration. Should we control for that?

Truth

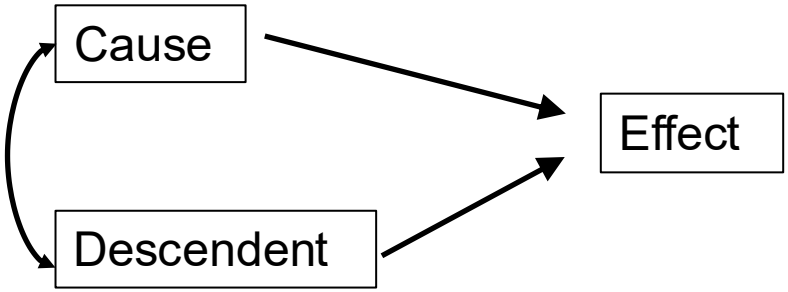


2. The Descendant Problem

Causal Model



Regression



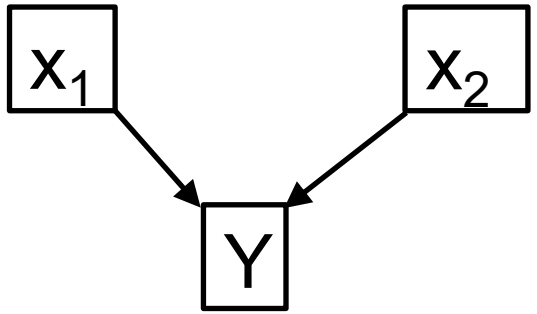
- **Descendant still partially blocks cause**
- **Masks effect of cause – many possibilities**

Elemental paths in causal graphs (and Why to use SEM)

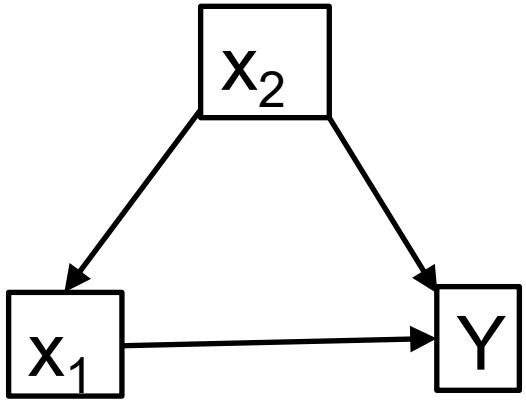
1. The Chain/Pipe

2. The Descendant

3. The Collider



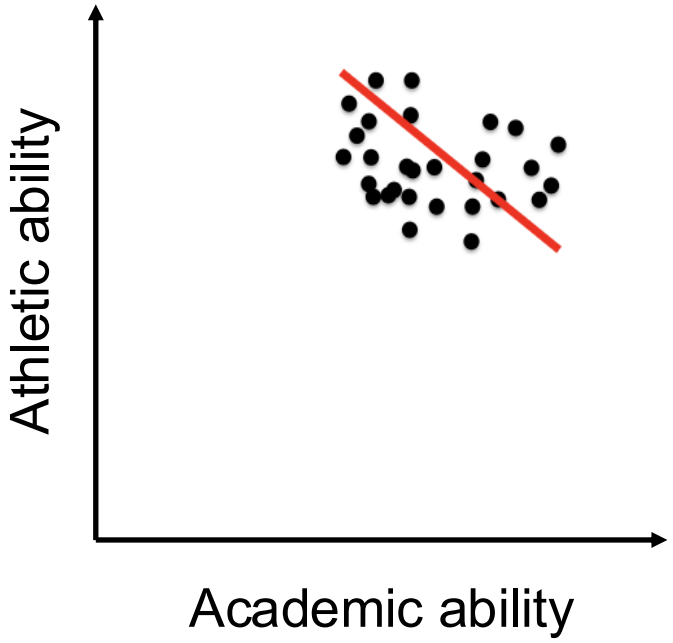
4. The Fork



A Collision of Sampling and Regression

Consider the following:

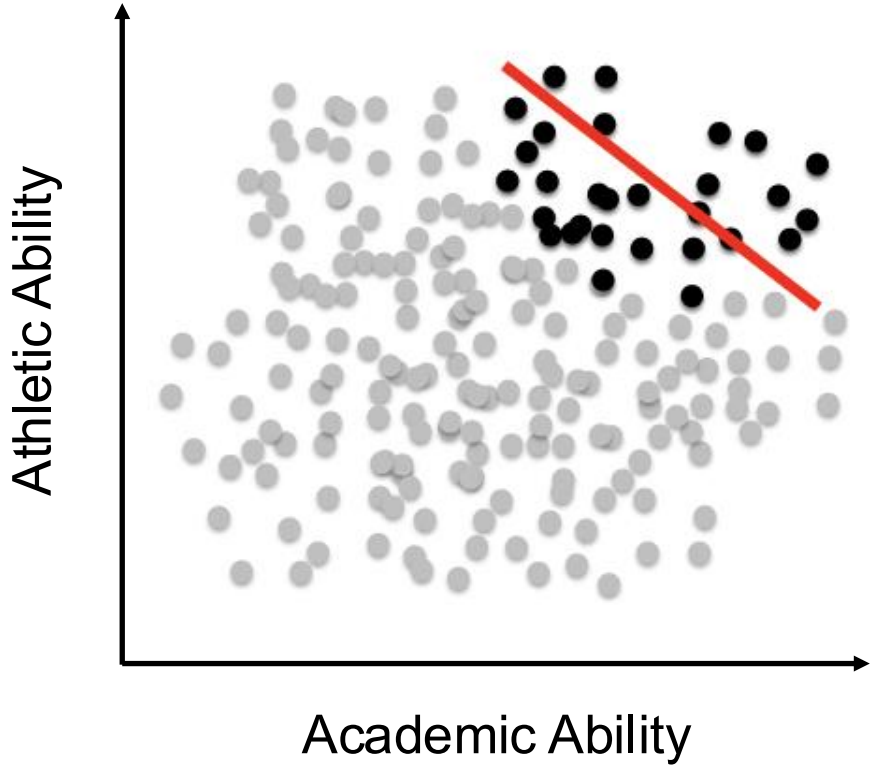
You are interested to see if academic ability and athletic ability are correlated. So, you sample students at your collage.



A Collision of Sampling and Regression

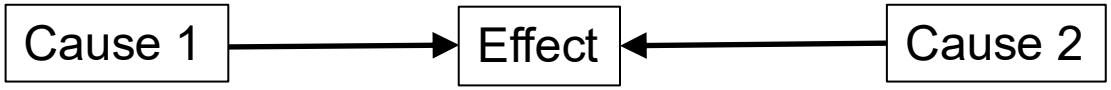
But how do you get into college?



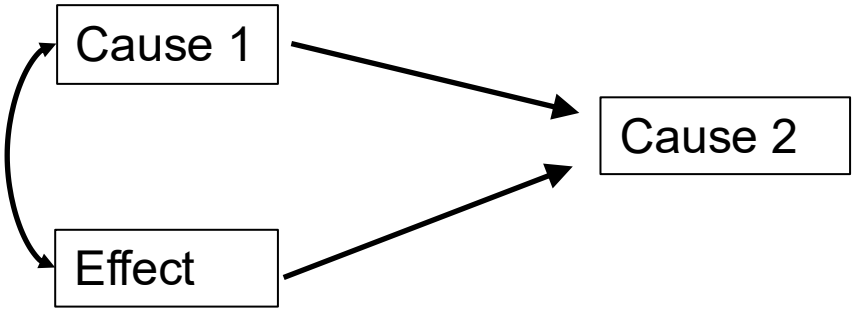


The Collider

Causal Model



Regression



- **Ignoring Effect, Cause 1 and Cause 2 NOT associated**
- **Conditioned by Effect, Cause 1 and Cause 2 associated**
- **Creates spurious correlations**

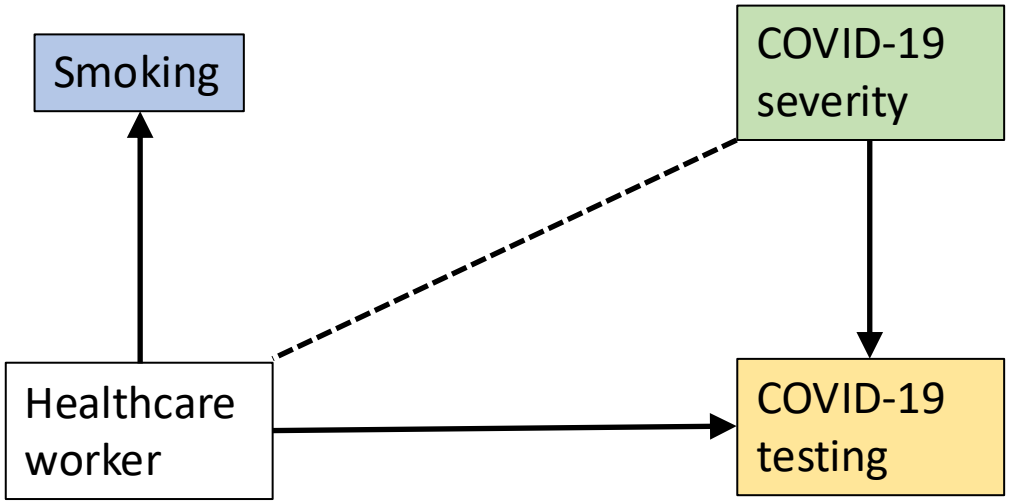
Collider Bias

Preliminary studies suggested that

a) Healthcare workers were likely to get less severe Covid infections

b) Covid infection was lower among smokers

WHY?



Griffith et al. 2020,
https://doi.org/10.1038/s41467-020-19478-2

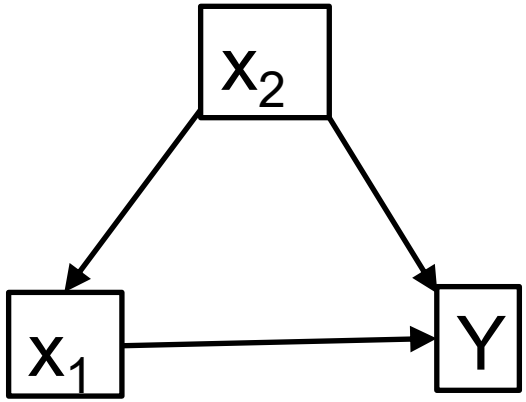
Elemental paths in causal graphs (and Why to use SEM)

1. The Chain/Pipe

2. The Descendant

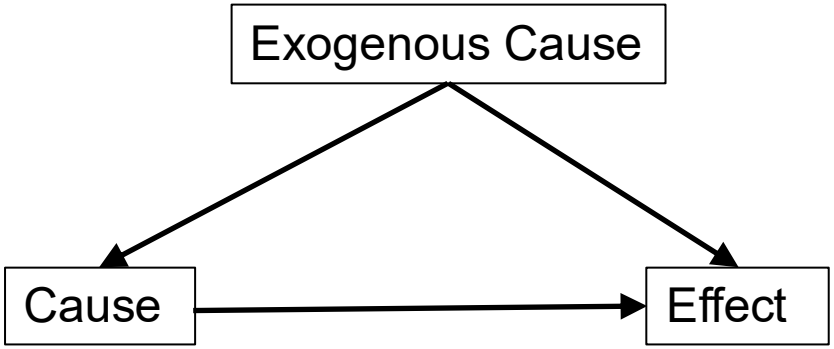
3. The Collider

4. The Fork

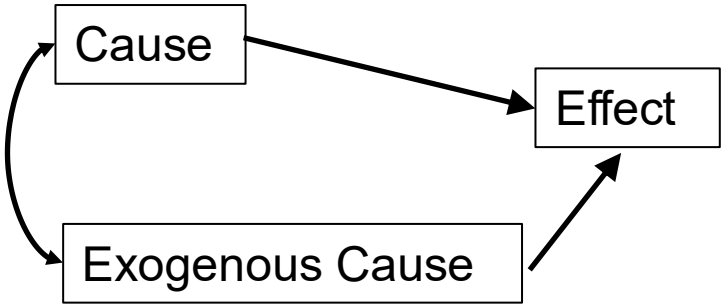


4. The Fork

Causal Model

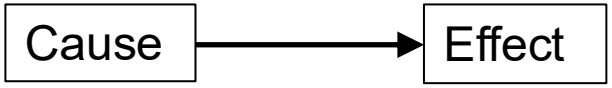


Regression 1



- **Conditioned by Exogenous Cause, Cause and Effect NOT associated**
- **Nothing – this is great!**

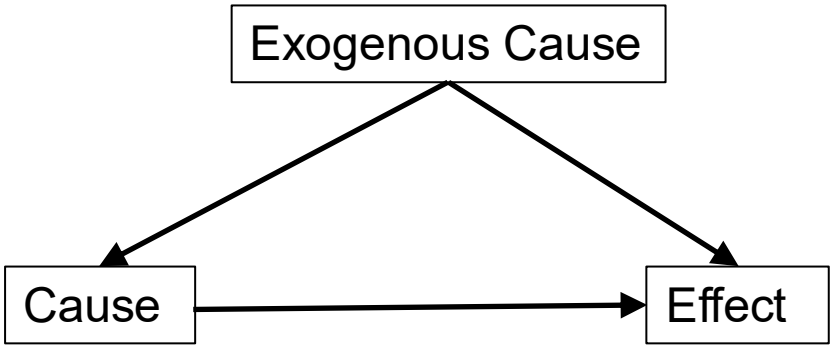
Regression 2



- **Ignoring Exogenous Cause, Cause and Effect associated**
- **The relationship is contaminated – results unreliable!**

The Bigger Problem: The Omitted Variable Bias Problem

Causal Model

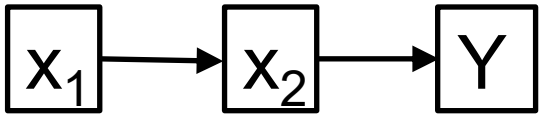




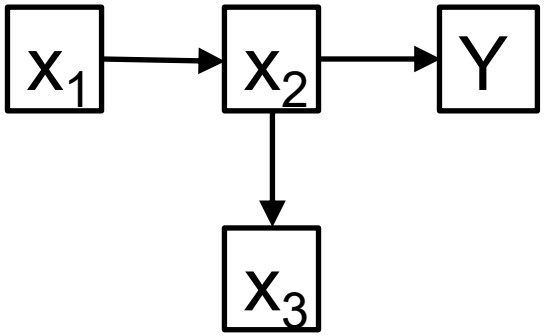
- **We assume that sampling means that omitted variables average to 0 – Omission produces downward bias in SE of coefficients.**
- **But, if omitted variables are correlated causally with a predictor, they likely are not averaged out.**
- **This will bias your estimates – You will not know in what direction.**

Elemental paths in causal graphs (and Why to use SEM)

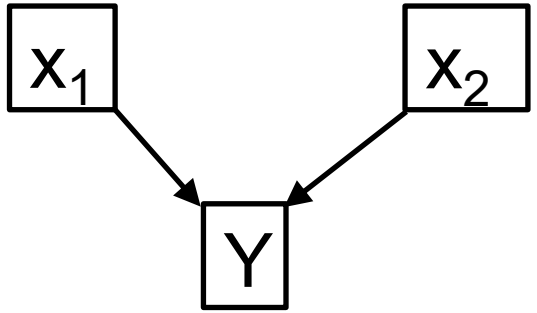
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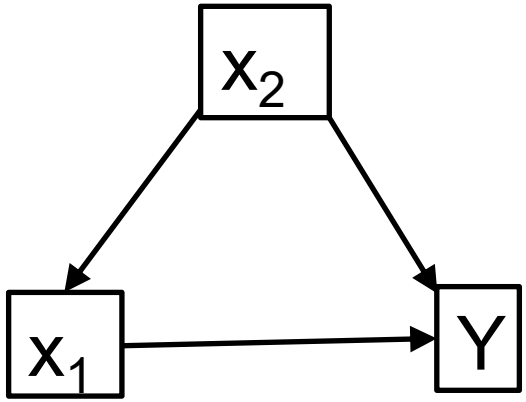
2. The Descendant



3. The Collider



4. The Fork



You are now Armed and Dangerous



Translating between the languages of Causation and Probability Theory

- How to replace the notion of physically controlling a variable in a causal system with the notion of a statistical control in a causal system.

Language of science
Concepts of cause - effect, direct and indirect causes, causal independence, etc.

→

Language of graph theory
Concepts of direct acyclic graphs (DAGs), paths, ancestors, dependents, etc.

D-separation →

Languages of probability theory
Concepts dependence, independence, conditional distributions, marginal distribution.....

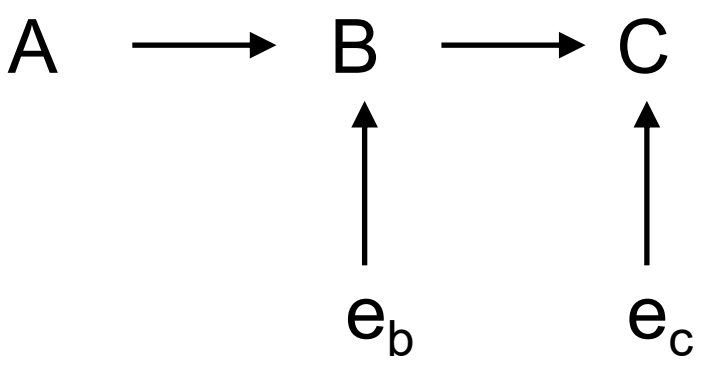


Causality: Models, Reasoning, and Inference (2000)

The Book of Why: The New Science of Cause and Effect (2018)

Expressing Causal Hypotheses using DAGs (Directed Acyclic Graphs)

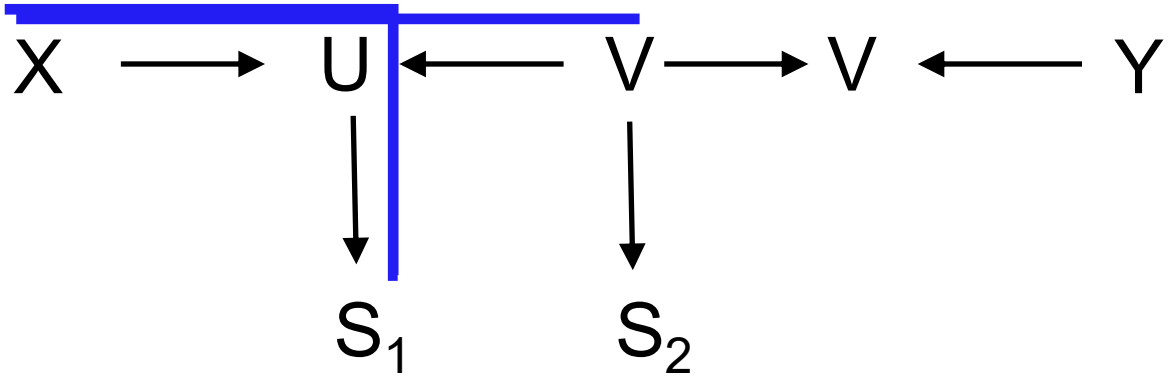
- Variables A, e_b and e_c are casually independent of each other.
- A and e_b each directly cause changes in variable B.
- Variables B and e_c are also causally independent of each other and each directly causes changes in variable C.



Important: the mathematical statement, X→Y in a DAG DOES NOT mean X is correlated with Y. It means that "A change in X will provoke a change in Y even when all other variables in the DAG are prevented from changing but a change in Y will not provoke a change in X"

The same causal hypothesis expresses as a DAG

D-separation is a mathematical operation on a DAG

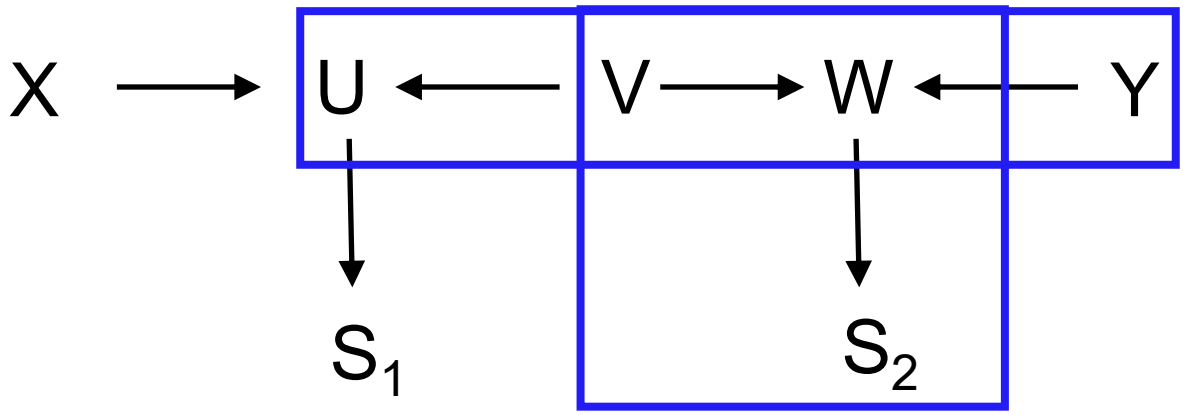


X →	U →	S_1	Directed path	Non-collider (U)
X →	U ←	V	(Undirected) path	Collider (U)

Natural state of a variable along a path:

- **Collider:** blocks the flow of statistical information through it.
- **Non-collider:** allows the flow of statistical information through it.

Let's do a Test



Is there a directed path from **V** to $\mathbf{S_2}$?

Is there an undirected path from **V** to $\mathbf{S_2}$?

Is **W** a collider or non-collider along the undirected path between **V** to **S2**

Is there a directed path from **U** to **Y** ?

Is **W** a collider or a non-collider along the undirected path between **U** to **Y** ?

D-separation is a Mathematical Operation on a DAG

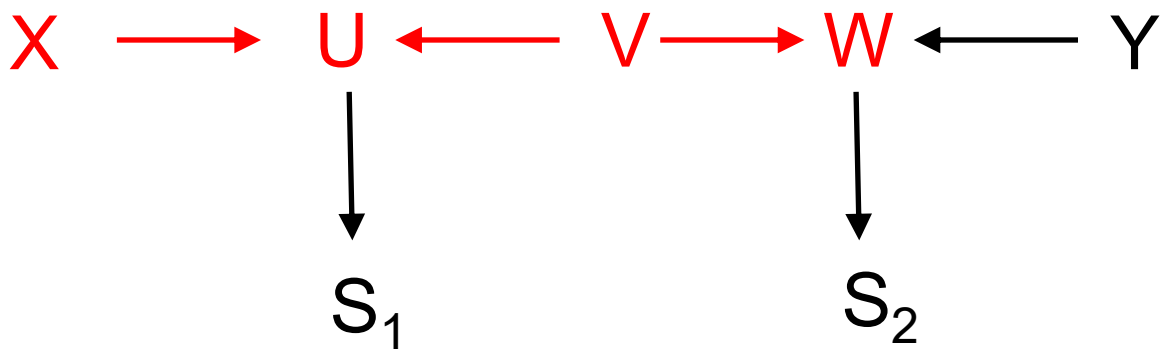
Natural state of a variable along a path:

- **Collider:** blocks the flow of statistical information through it.
- **Non-collider:** allows the flow of statistical information through it.

- Conditioning on a **non-colder** reverses its natural stat and **blocks** the flow of statistical information through it.

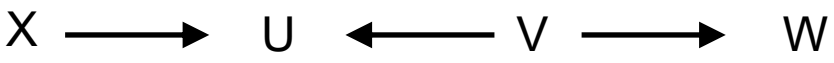
- Conditioning on a **collider or** on the **descendent of a collider reverses** its natural state and **allows** the flow of statistical information through it.

An example of a d-Separation claim



Is **X** d-separated from **W** given **U** ? **X ⫫ W {U}** ?

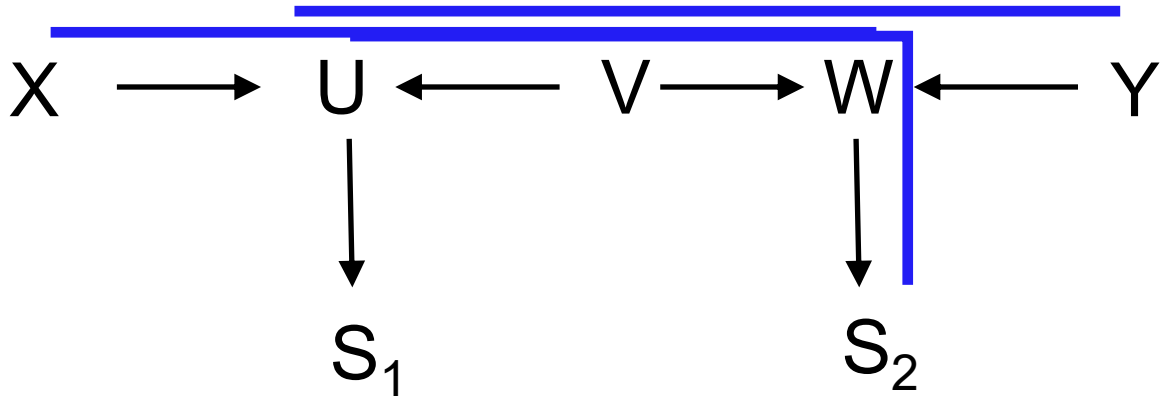
Step 1: Write down all (undirected) paths between **X** and **W**



Step 2: Now check the conditional set **{U}**

Concussion: X is not d-separated from **W**, given (conditional on) **U**

An example of a d-Separation claim



Is **U** d-separated from $\mathbf{S_2}$ given **V** ?

$\mathbf{U \perp\!\!\!\perp S_2 \{V\}}$?

Y ⫫ U {S2} ?

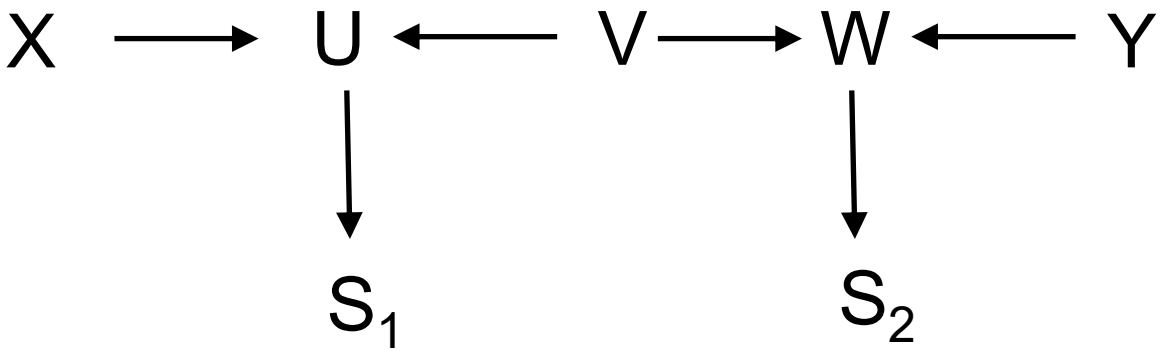
$$\frac{\mathbf{V!}}{\mathbf{(V-2)!\ 2!}}\ \mathbf{2^{V-2}}$$

X ⫫ W {S1, V} ?

Basis set

- You only need to test those in the basis test

Practicing to master the notion of d-Separation




```
library(ggm)

my.dag<-DAG(U~ X + V, S1 ~ U, W ~ V + Y, S2 ~W)

# evaluating a d-separation claim

# U_||_S2|{X,W}
dSep(amat=my.dag,first="U",second="S2",cond=c("X","W"))
```

Open Script_2 and run codes under “### exercise 1

Why is the d-Sep test performed on a DAG?

- The d-sep test is designed to tell us whether we should retain the proposed DAG or not. If the DAG is rejected, we stop there and conclude that the data were not generated according to our hypothesised causal model.

The d-sep test of A DAG

The null hypothesis:

(i). The observed data were generated by the hypothesized DAG.

(ii). Each of the predicted d-separation claims in the DAG are true.

If the null hypothesis is true, the the *C* statistic will not be unusually large and the null probability will not be unusually small

- If the null probability is larger than your chosen significance level (alpha = 0.05) then we DO NOT reject the null hypothesis (and your DAG).

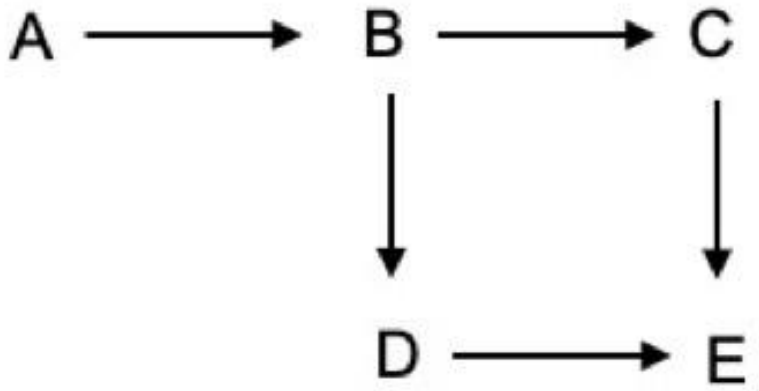
E.g.,

- Root Mean Square Error of Approximation (RMSEA)
- Comparative Fit Index (CFI)
- Standardized Root Mean Square Residual (SRMR)

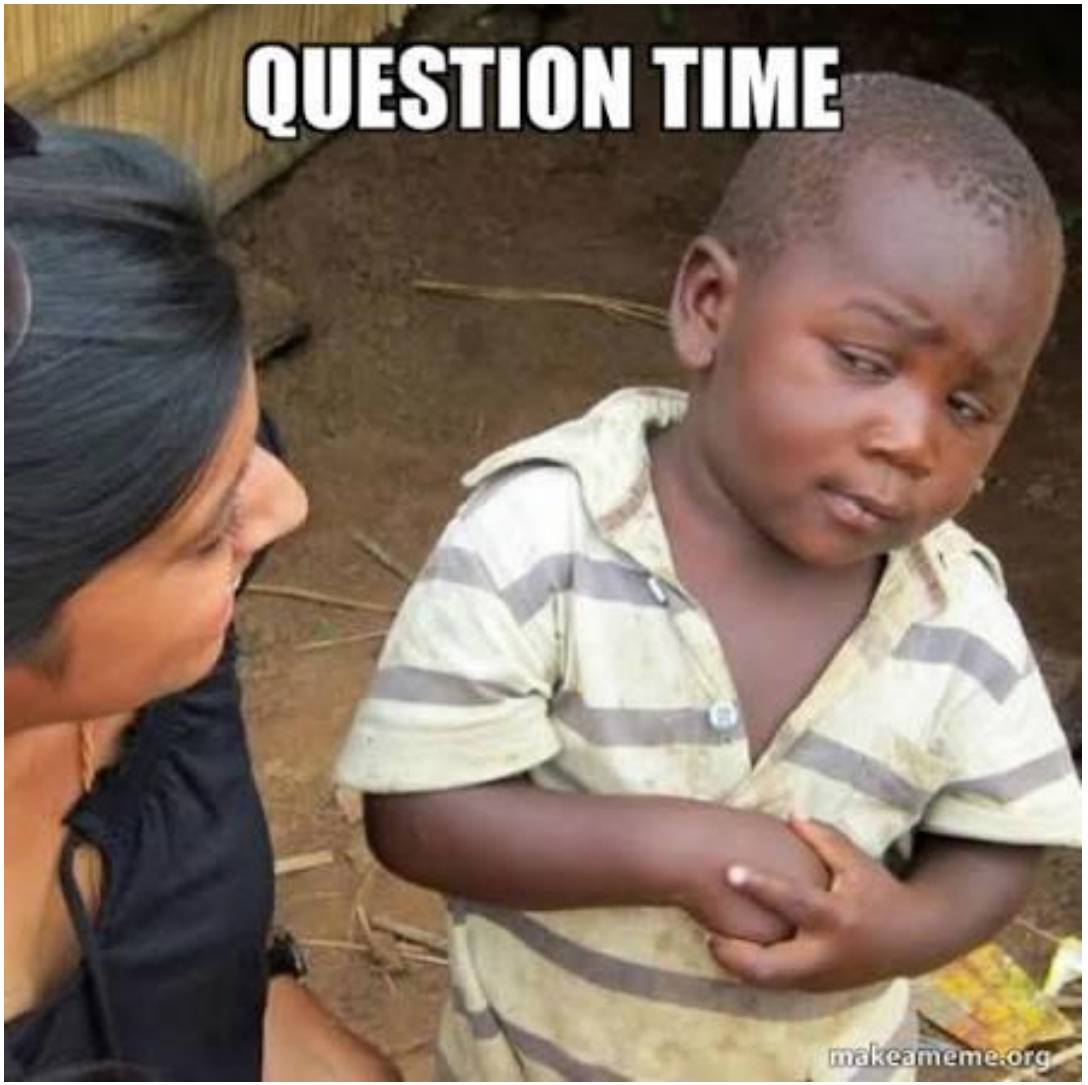
Scenario:
Conservation scientists suggest that bee abundance in Singapore has declined due to urbanisation. As a solution, they propose that increasing the number of urban parks with greenery could help enhance bee populations.

Research Question:
Can bee abundance be increased by adding more urban parks to Singapore's landscape?

Causal hypotheses:
- An increase in the number of urban parks (A) enhances plant cover (B) within Singapore's urban landscape.
- Greater plant cover (B) positively influences both floral richness (C) and floral abundance (D).
- In turn, higher floral richness (C) and floral abundance (D) contribute to increased bee abundance (E).



Open Script_2 and run codes under "### exercise 2


QUESTION TIME
makeameme.org

Structural Equation Modeling

Structural

There is hypothesized underlying structure to nature (a cause and an effect)…

Equation

…that can be translated to a series of mathematical equations…

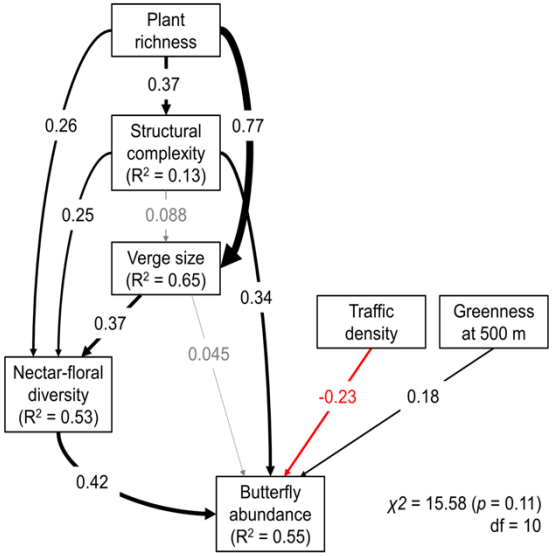
Modeling

…which can be modelled against data to support or refute the proposed structure

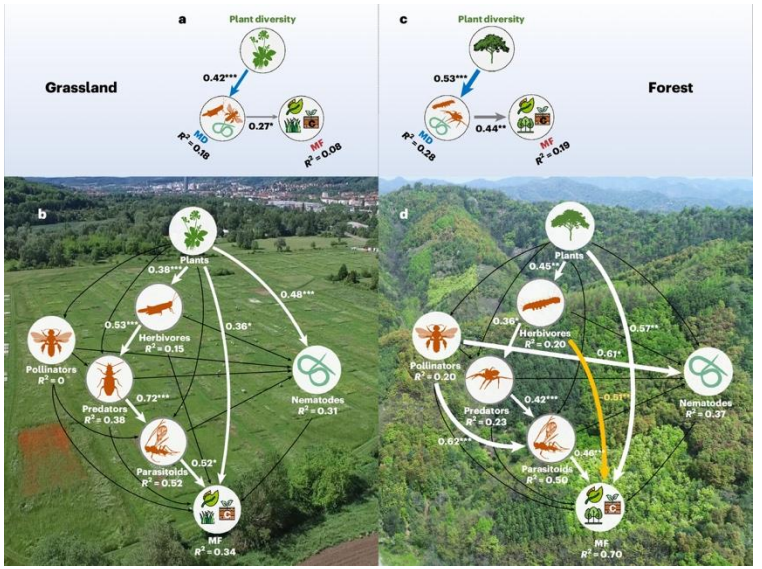
S **E** **M**

What is Structural Equation Modelling?

- Can be thought of as a series of equations united in a single causal network.
- Can investigate the flow of effects through a system (i.e., simultaneously estimating both direct and indirect effects of variables on each other).
- Can be used to test an a priori hypothesized path model or to develop an exploratory model.

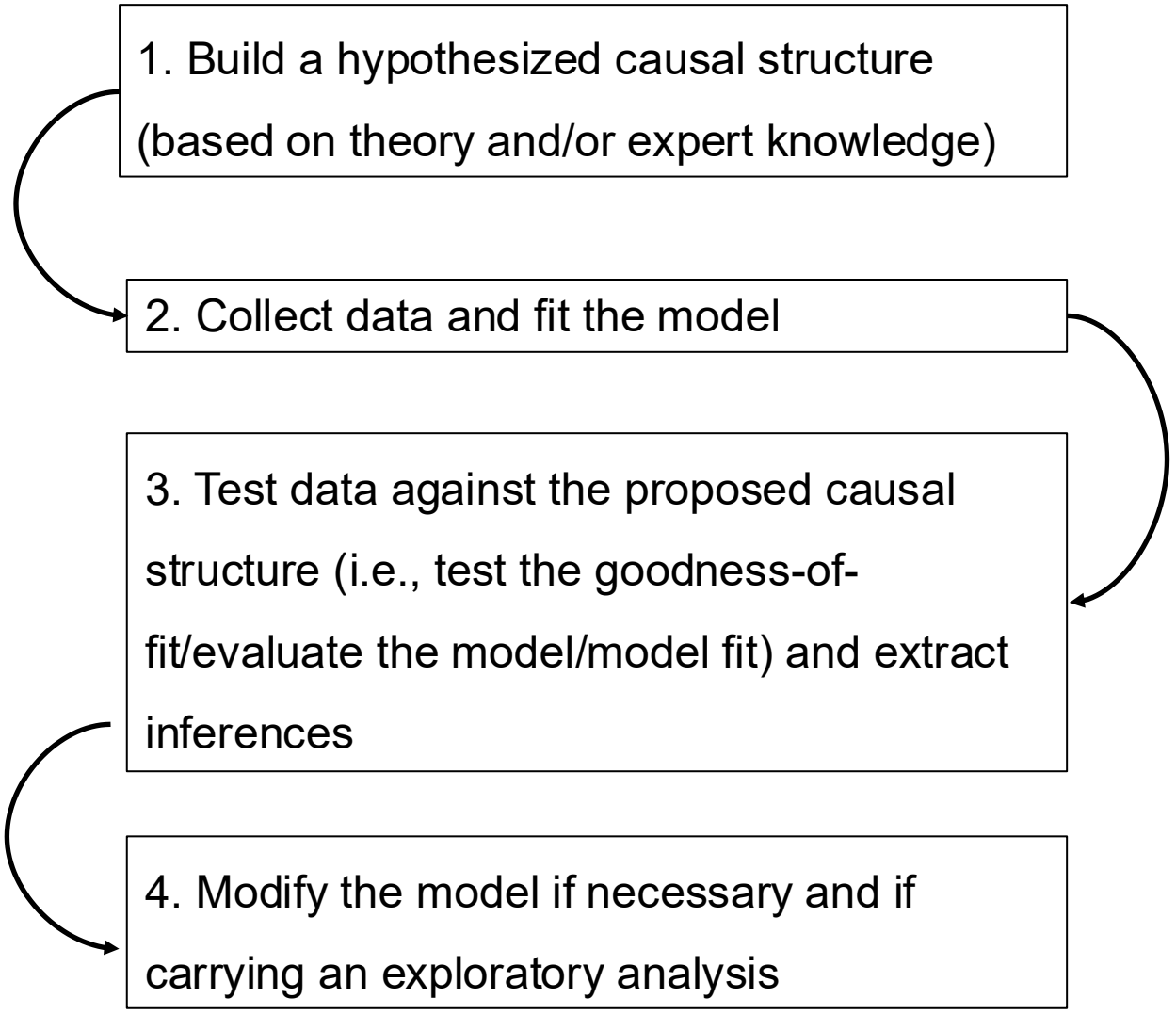


Priyadarshana et al., 2025, 10.1016/j.landurbplan.2025.105335



Li et al., 2024, https://doi.org/10.1038/s41559-024-02517-2

SEM analysis steps



Global and Local estimation SEMs

Global estimation SEM

- Considers the model as a whole
- Compares observed covariances to model covariances
- A linear modeling technique
- **lavaan** R-package

Local estimation SEM

- Breaks the model in sub-models
- Evaluates the importance of missing paths for the model
- Allows generalized mixed models
- **piecewiseSEM** R-package

Advantages and Limitations of SEM

- Study the direct and indirect effects of explanatory variables.
- It does not assume that only the dependent variable is subject to measurement error.
- Evaluates the whole model.
- Can be used to test latent variables (cannot be observed or measured).
- Large sample size.
- The method is continuously developed.

Let’s go through a real example

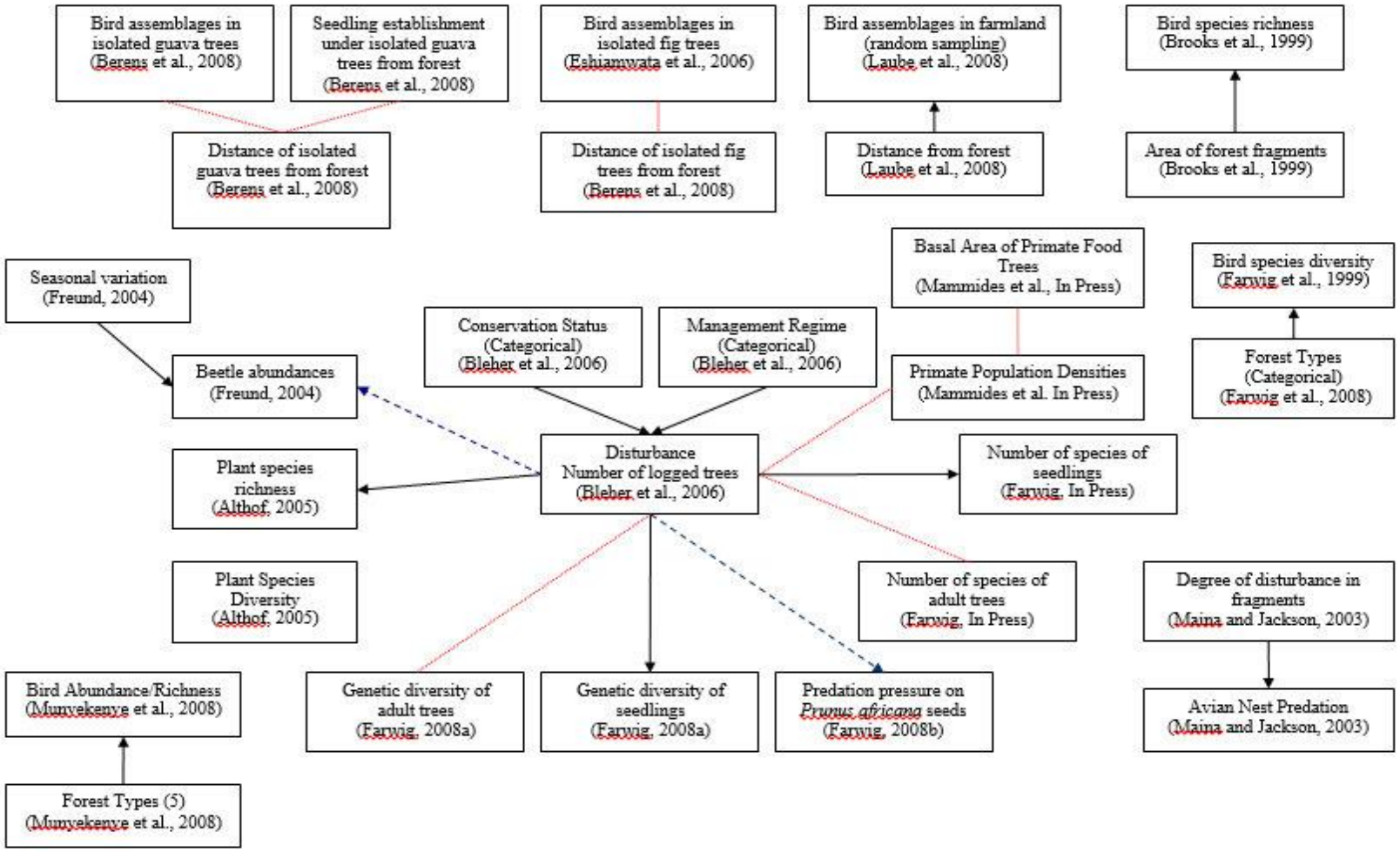
The indirect effects of habitat disturbance on the bird communities in a tropical African forest

Mammides et al., 2015

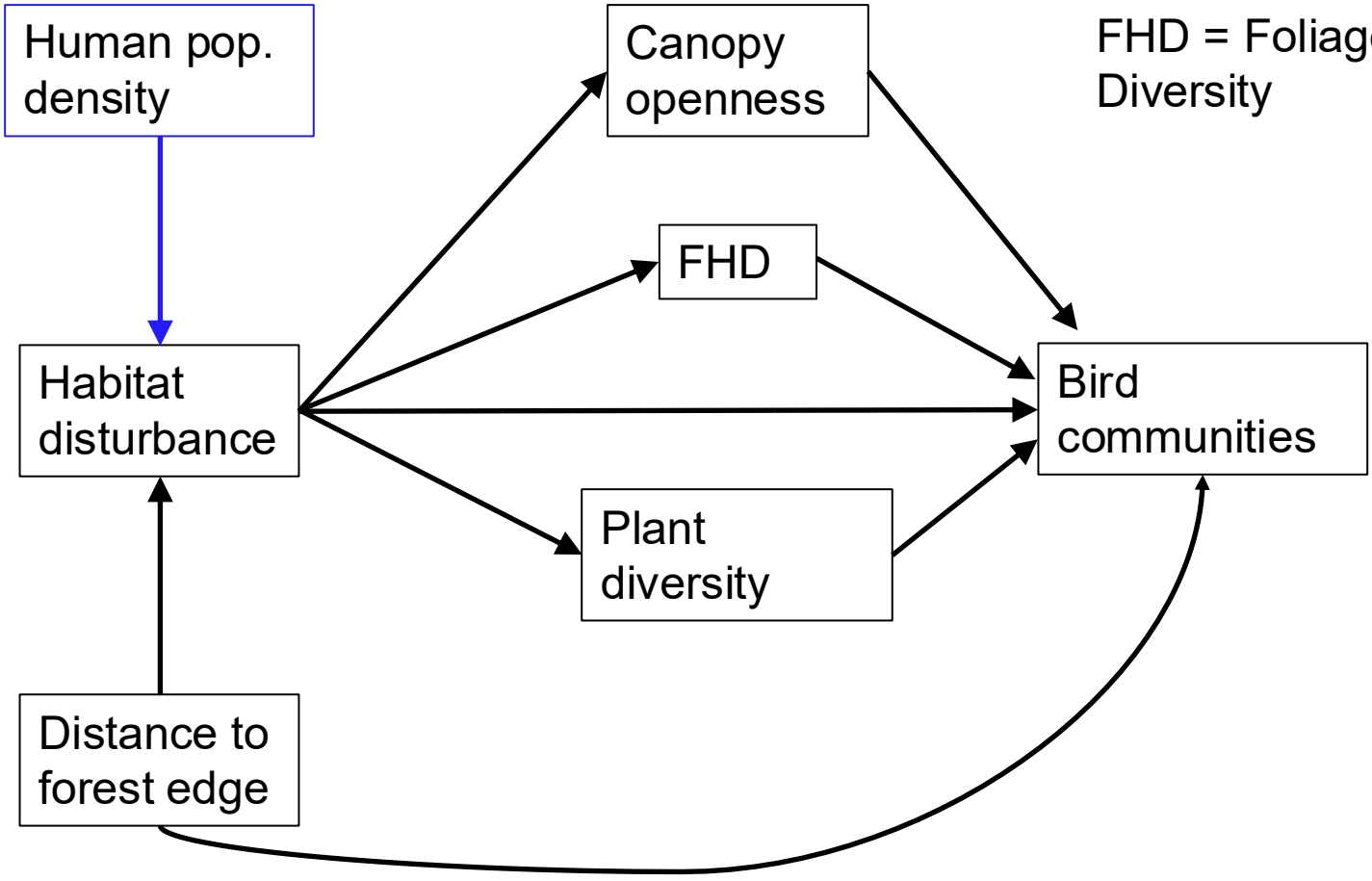
Biodiversity Conservation

https://doi.org/10.1007/s10531-015-1001-x

Reviewed the literature

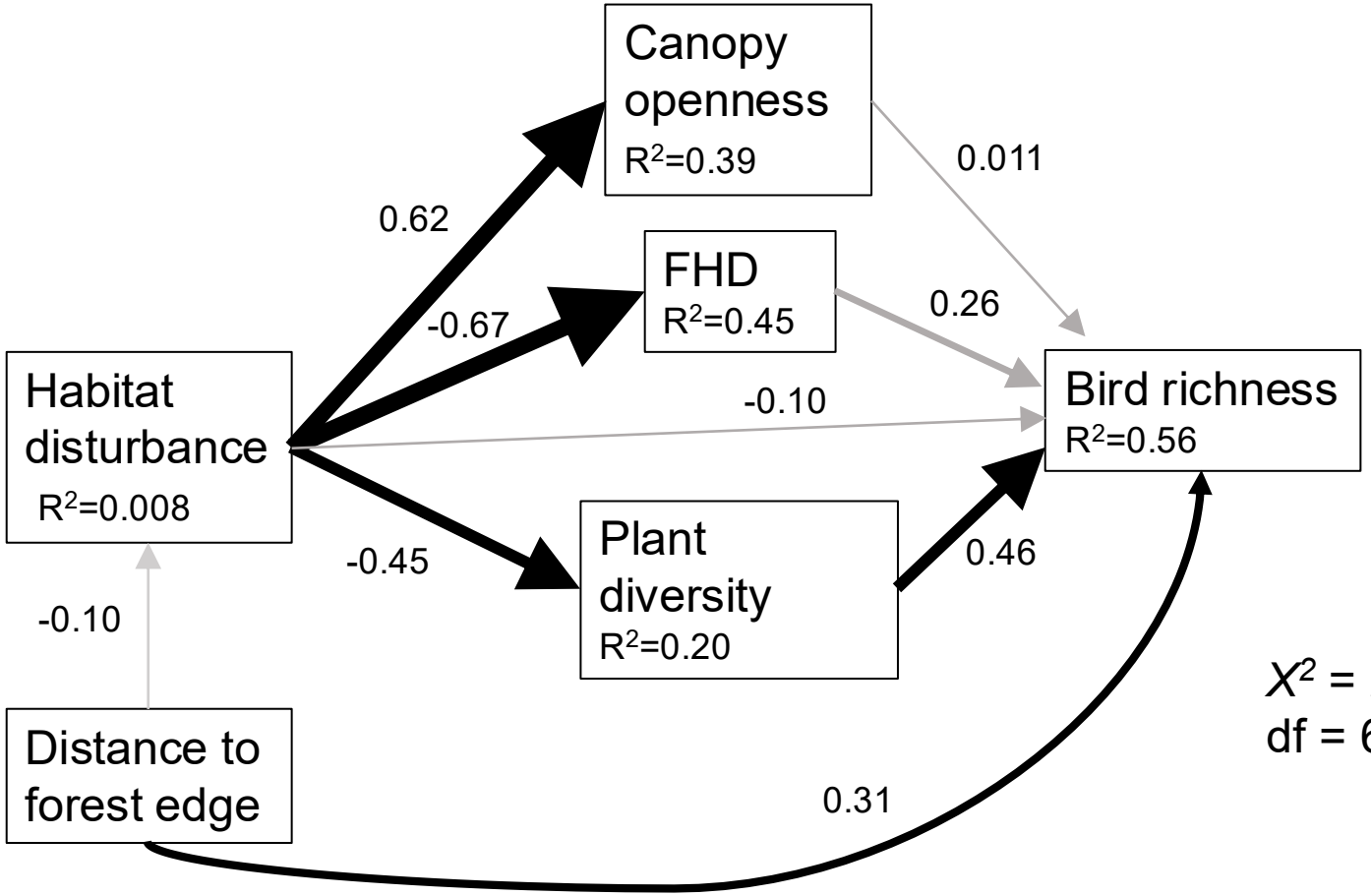


Proposed casual model/DAG



Open Script_3 and run codes for this analysis

Results



X^2 = 2.76 (p = 0.89)
df = 6

Bird Assemblages	Indirect effects	Direct effects	Total effects
Overall Richness	-0.37	-0.1	-0.47

HD → CO → BR = 0.62 x 0.011 = 0.00682
HD → FHD → BR = -0.67 x 0.26 = -0.1742
HD → PD → BR = -0.45 x 0.46 = -0.207

0.00682 +(-0.1742)+(-0.207) = -0.37

Think Critically!

- SEM is a tool, it is up to the user to employ it thoughtfully.
- Do not rely on the statistics as much as what you know about biology and ecology.
- If the answer does not make sense, assume the model or test is wrong!

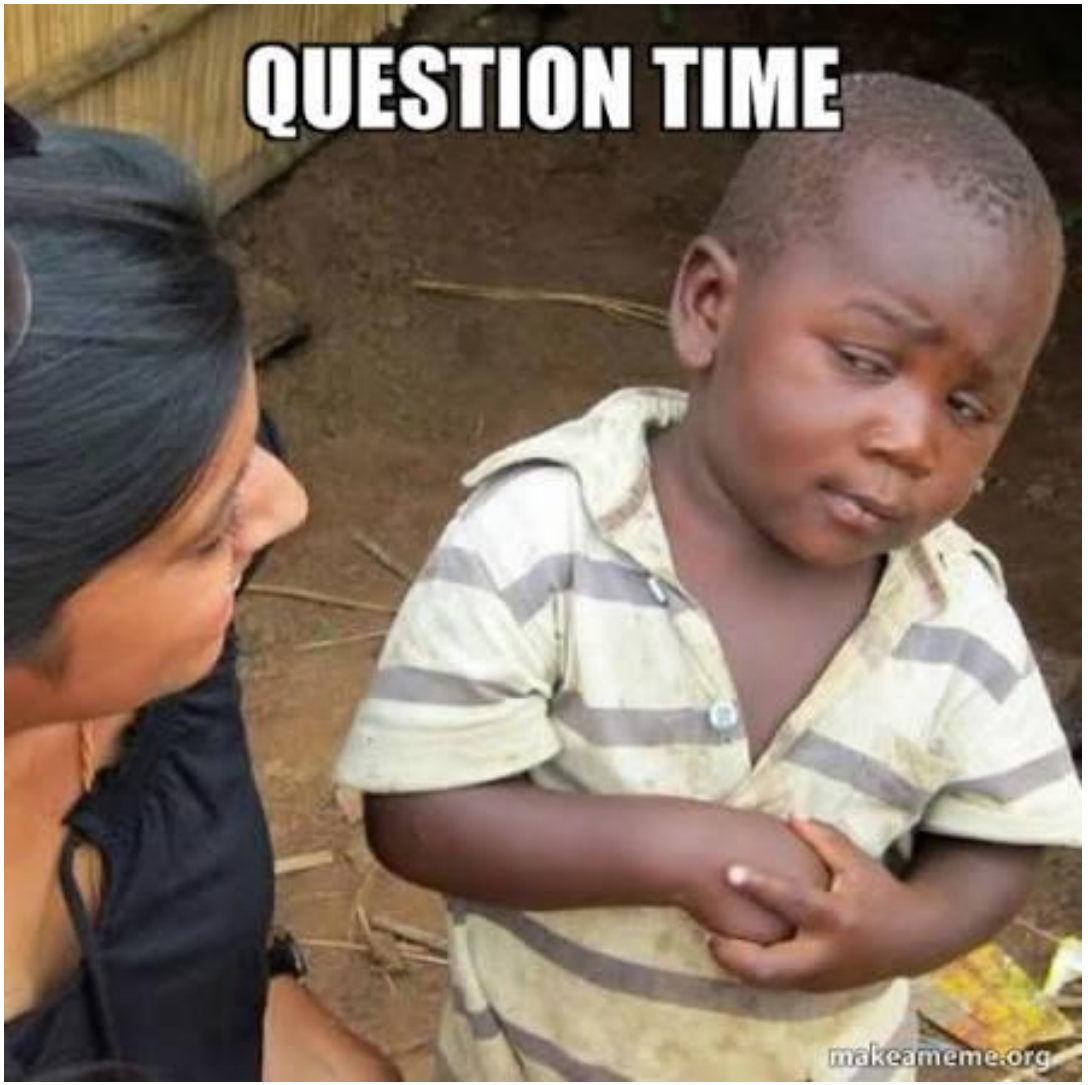
My Advice!

- Start by drawing a causal diagram (DAG) that reflects your research question(s).
- Build your DAG based on evidence from previous studies, your understanding of the study system and underlying ecological mechanisms, and expert knowledge.
- “No causes in, no causes out!” — Nancy Cartwright

Remember!

- Your DAG should represent your causal assumptions, not simply the statistical relationships observed in your data.

Priyadarshana & Slade 2026,
https://doi.org/10.1016/j.cois.2026.101538


QUESTION TIME
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